

## Overview

CH32X035 is an industrial-grade microcontroller based on the QingKe RISC-V core. CH32X035 has built-in USB and PD PHY, supports USB Host and USB Device functions, PDUSB and Type-C fast charging functions, built-in programmable protocol I/O controller, provides 2 groups of OPA, 3 groups of CMP, 4 groups of USART, I2C, SPI, multiple timers, 12-bit ADC, 14-channel touch-key and other rich peripheral resources.

## Features

- **Core**
  - QingKe 32-bit RISC-V4C core
  - Support RV32IMAC instruction set and self-extending instructions
  - Fast programmable interrupt controller + hardware interrupt stack
  - Branch prediction, conflict handling mechanism
  - Single-cycle multiplication, hardware division
- **Memory**
  - 20KB volatile data storage area SRAM
  - 62KB program memory area CodeFlash
  - 3328B system memory area SystemFLASH
  - 256B system non-volatile configuration information memory area
  - 256B user-defined information storage area
- **Power management and low-power:**
  - System power supply  $V_{DD}$  rated voltage: 3.3V or 5V
  - Low power modes: Sleep, Stop, Standby
- **Clock & Reset**
  - Built-in 48MHz RC oscillator
  - Power on/down reset, programmable voltage detector
- **8-channel general-purpose DMA controller**
  - 8 channels, support ring buffer management
  - Support TIMx/ADC/USART/I2C/SPI
- **Programmable Protocol I/O Controller PIOC:**
  - Programmable, supports a wide range of 1-wire interfaces, 2-wire interfaces
- **2-group OPA/PGA/CMP:**
  - Multiple input channels, selectable multi-step gain
  - 2 output channels each, optional ADC pins
- **3-group analog voltage comparator CMP:**
  - 2 input channels each, optional common reference voltage pin
  - Output to I/O or internal direct trigger TIM2
- **12-bit ADC**
  - Analog input range:  $GND \sim V_{DD}$
  - 14-channel external signal + 1-channel internal signal channel
- **14-channel touch-key detection**
- **Multiple timers**
  - 2×16-bit advanced-control timers, with dead zone control and emergency brake; can offer PWM complementary output for motor control
  - 1×16-bit general-purpose timers, provide input capture/output comparison/PWM
  - 2 watchdog timers (independent watchdog and window watchdog)
  - SysTick: 64-bit counter
- **4-group USART: Support LIN and ISO7816**
- **1×I2C interface: Support SMBus/PMBus**
- **1×SPI interface**
- **USB2.0 full-speed controller and PHY:**
  - Support USB Host and USB Device
- **USB PD and Type-C controller and PHY**
- **Fast GPIO port**
  - 60 I/O ports, with 24 external interrupts
- **Security features: Chip unique ID**
- **Debug mode: 2-wire serial debug interface (SDI)**
- **Package: LQFP, QFN, QSOP, TSSOP**

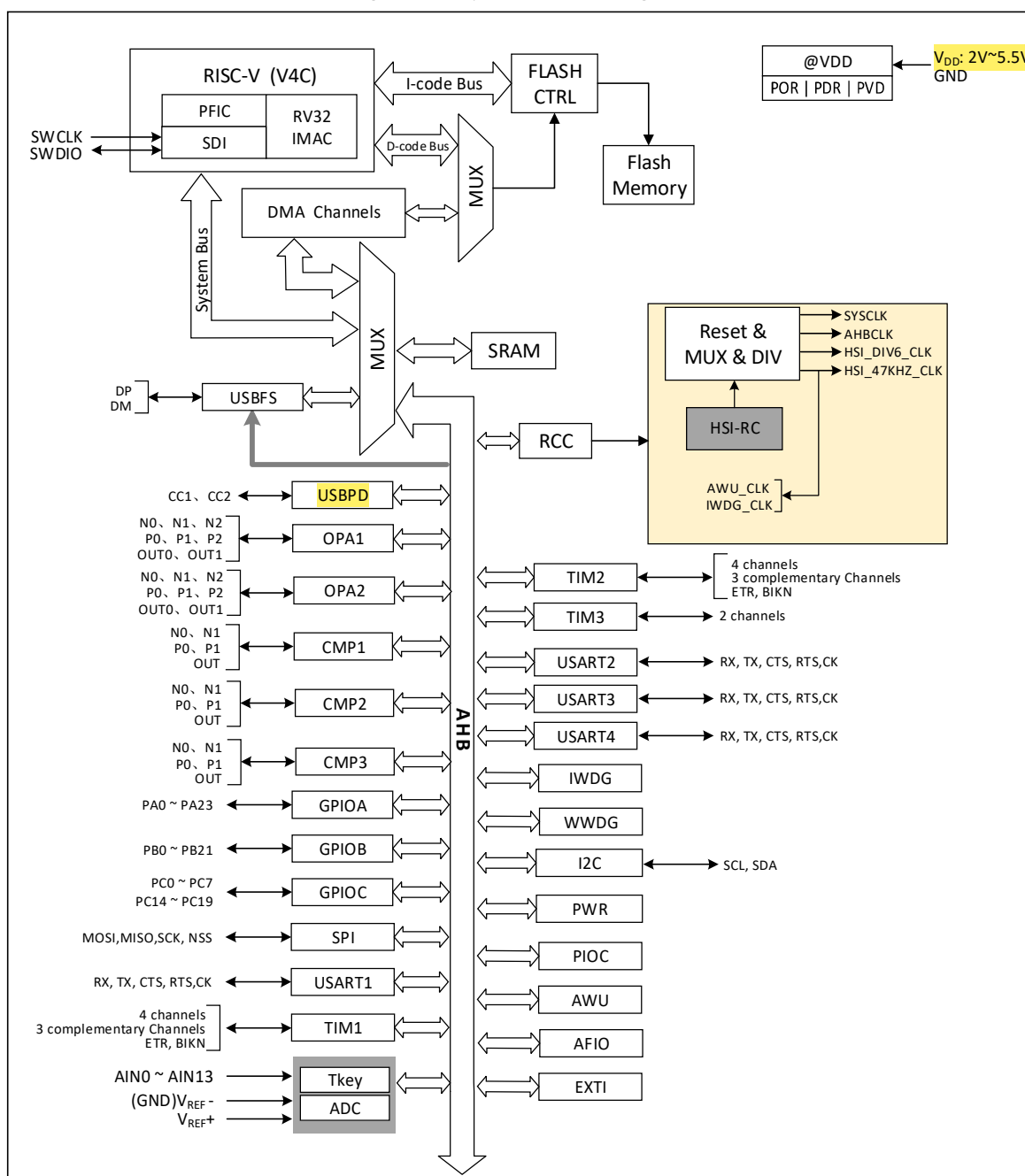
| Model        | Flash | RAM | GPIO | Advanced-<br>control<br>timer | General-<br>purpose<br>timer | Serial<br>port | Watchdog | PDUSB       |               |                                 | ADC  | OPA | CMP | Capacitive<br>Touchkey | SPI | PIOC<br>1-wire<br>interface | Package form |
|--------------|-------|-----|------|-------------------------------|------------------------------|----------------|----------|-------------|---------------|---------------------------------|------|-----|-----|------------------------|-----|-----------------------------|--------------|
|              |       |     |      |                               |                              |                |          | USB<br>Host | USB<br>Device | Type-C<br>Source<br>Sink<br>DRP |      |     |     |                        |     |                             |              |
| CH32X035R8T6 | 62K   | 20K | 60   | 2                             | 1                            | 4              | 2        | √           | √             | √                               | 14+1 | 2   | 3   | 14                     | √   | √                           | LQFP64M      |
| CH32X035C8T6 | 62K   | 20K | 46   | 2                             | 1                            | 4              | 2        | √           | √             | √                               | 10+1 | 2   | 3   | 10                     | √   | √                           | LQFP48       |
| CH32X035G8U6 | 62K   | 20K | 27   | 2                             | 1                            | 4              | 2        | √           | √             | √                               | 10+1 | 2   | 1   | 10                     | √   | √                           | QFN28        |
| CH32X035G8R6 | 62K   | 20K | 26   | 2                             | 1                            | 4              | 2        | √           | √             | √                               | 11+1 | 2   | 3   | 11                     | √   | √                           | QSOP28       |
| CH32X035F8U6 | 62K   | 20K | 19   | 2                             | 1                            | 3              | 2        | -           | √             | √                               | 10+1 | 2   | -   | 10                     | √   | √                           | QFN20        |
| CH32X035F7P6 | 62K   | 20K | 18   | 2                             | 1                            | 3              | 2        | -           | √             | √                               | 11+1 | 1   | 1   | 11                     | √   | √                           | TSSOP20      |
| CH32X033F8P6 | 62K   | 20K | 18   | 2                             | 1                            | 4              | 2        | -           | √             | -                               | 10+1 | 2   | 2   | 10                     | √   | √                           | TSSOP20      |

## Chapter 1 Specification Information

## 1.1 System Structure

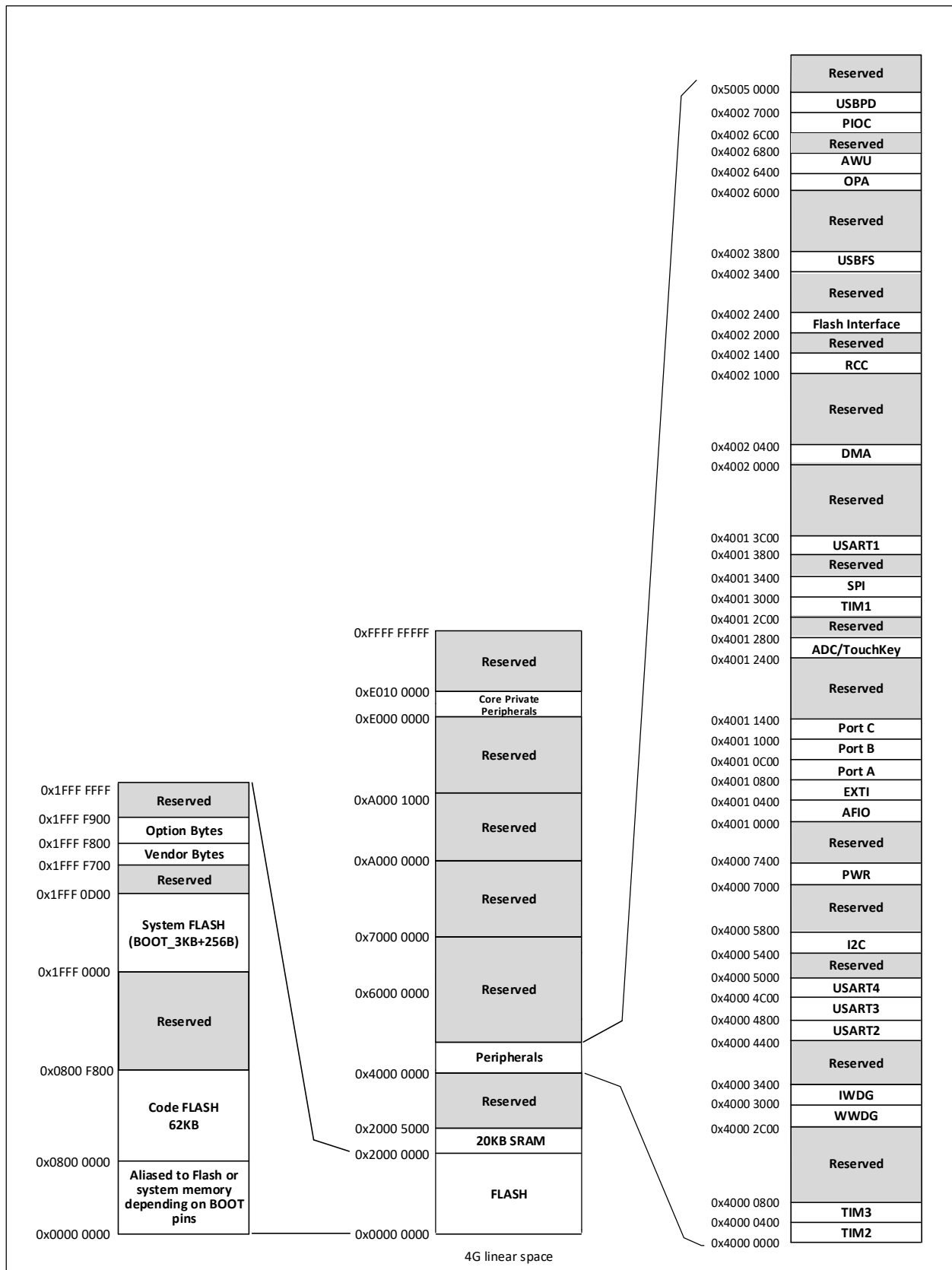
The microcontroller is designed on the basis of the RISC-V instruction set, and its architecture integrates the barley microprocessor core, arbitration unit, DMA module, SRAM storage and other components through multiple bus groups to achieve interaction. A general-purpose DMA controller is integrated to reduce the CPU load and improve access efficiency, and a multi-level clock management mechanism is applied to reduce the power consumption of peripherals. The following diagram shows the overall internal architecture of the series chip.

Figure 1-1 System Block Diagram



## 1.2 Memory Map

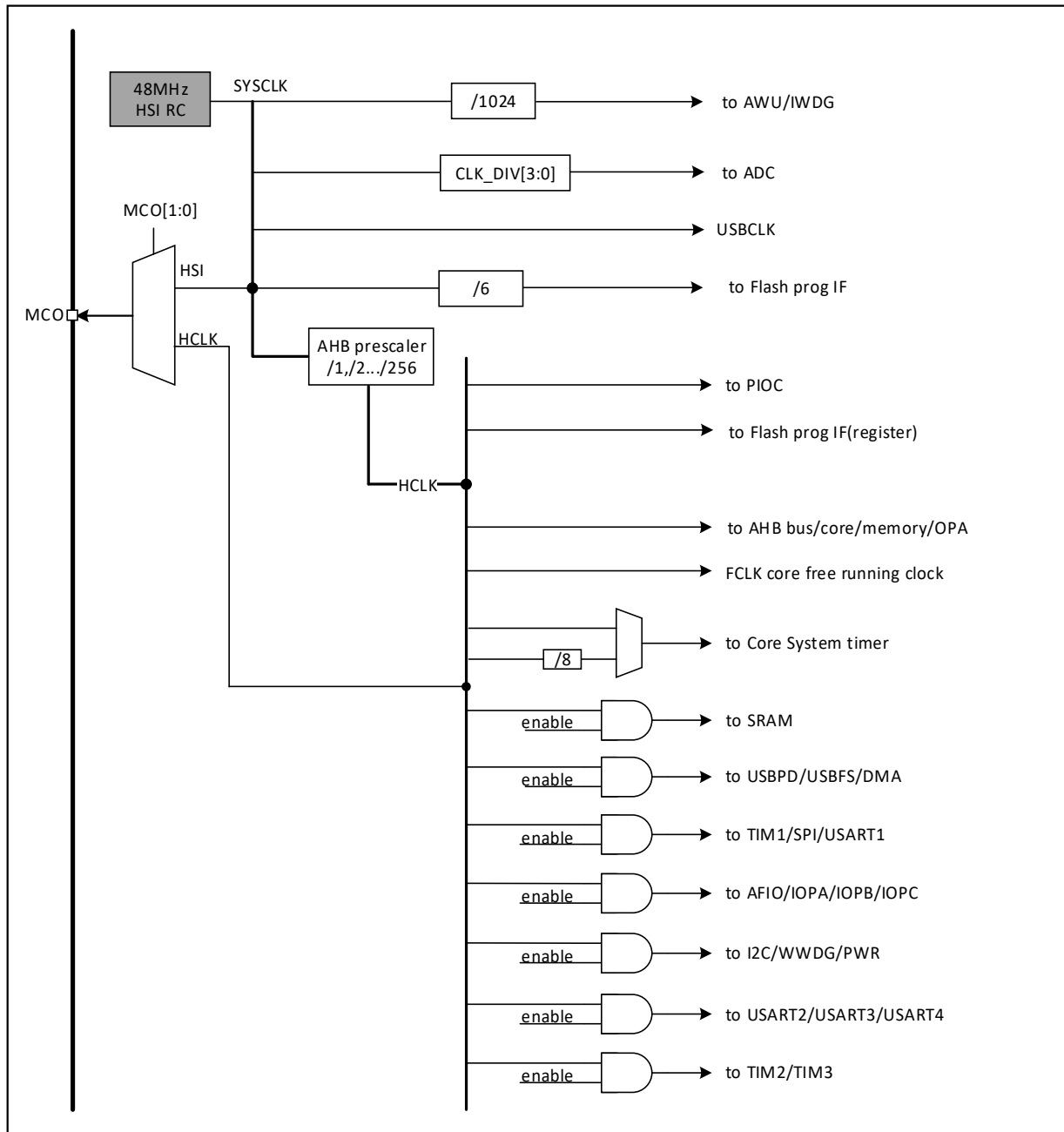
Figure 1-2 Memory address map



## 1.3 Clock Tree

System clock source: Internal high frequency RC oscillator (HSI)

Figure 1-3 Clock Tree Block Diagram



## 1.4 Functional Description

### 1.4.1 RISC-V4C Processor

RISC-V4C supports the IMAC subset of the RISC-V instruction set. The processor is managed internally in a modular fashion and contains units such as a programmable fast interrupt controller (PFIC), memory protection, branch prediction mode and extended instruction support. Externally multiple buses are connected to external unit modules, enabling interaction between external function modules and the core.

The processor with its minimal instruction set, multiple operating modes, and modular custom extensions can be flexibly applied to different scenarios of microcontroller design, such as small area low-power embedded scenarios, high performance application operating system scenarios, etc.

- Support machine and user privileged modes
- Programmable Fast Interrupt Controller (PFIC)
- Multi-level hardware interrupt stack
- 2-wire serial debug interface (SDI)
- Standard memory protection design
- Static or dynamic branch prediction, efficient jumping, conflict detection mechanisms
- Custom extension instructions

### 1.4.2 Programmable Protocol I/O Controller (PIOC)

Programmable protocol I/O controller is based on a single clock cycle dedicated compact instruction set RISC core running at system mains frequency with 2K instruction program ROM and 49 SFR registers and PWM timer/counter, supporting protocol control of 2 I/O pins.

- Multiplex 4K bytes of system SRAM as a 2K word capacity program ROM, supporting program pause and dynamic loading.
- Provide 33 bytes of 1 register each in bidirectional and unidirectional mode, providing 6 levels of independent stacking.
- Support 1- and 2-wire interfaces with multiple protocol specifications by dynamically loading different protocol programs.

### 1.4.3 On-chip Memory

Built-in 20K bytes SRAM area for data storage and data loss after power down. Among them, 4K can be used for PIOC.

Built-in 62K bytes program flash memory storage area (Code FLASH), i.e. user area, for user's application program and constant data storage.

Built-in 3328 bytes system storage area (System FLASH), i.e. BOOT area, for system boot program storage, built-in bootstrap loading program.

256-byte system non-volatile configuration information storage area, used for vendor configuration word storage, factory-cured, user cannot be modified.

256-byte user-defined information storage area for user-selected word storage.

### 1.4.4 Power Supply Scheme

$V_{DD} = 2 \sim 5.5V$ : supplies power to the I/O pins and internal regulator. (When using the ADC,  $V_{DD}$  should be no less than 2.5V.)

### 1.4.5 Power Supply Monitor

A power-on reset (POR)/power-down reset (PDR) circuit is integrated inside the chip, which is always active to ensure that the system operates when the power supply exceeds 2V; when  $V_{DD}$  is below the set threshold ( $V_{POR/PDR}$ ), it puts the device in reset without the need to use external reset circuitry.

In addition, the system has a programmable voltage monitor (PVD), which needs to be turned on by software, to compare the voltage magnitude of the  $V_{DD}$  supply with the set threshold  $V_{PVD}$ . Turning on the corresponding edge interrupt of the PVD allows you to receive an interrupt notification when  $V_{DD}$  falls to the PVD threshold or rises to the PVD threshold. Refer to Chapter 3 for  $V_{POR/PDR}$  and  $V_{PVD}$  values.

### 1.4.6 System Voltage Regulator LDO

After resetting, the system voltage regulator is automatically switched on. There are two modes of operation depending on the application mode.

- On mode: normal running operation, providing stable core power.
- Low-power mode: low-power operation of the regulator when the CPU is in Standby mode.

### 1.4.7 Low-power Mode

The system supports three low-power modes, which can be selected to achieve the best balance for conditions such as low-power consumption, short start-up times and multiple wake-up events.

- Sleep mode

In Sleep mode, only the CPU clock is stopped, but all peripheral clocks are powered normally and the peripherals are in working condition. This mode is the shallowest low-power mode, but the fastest wake-up can be achieved.

Exit condition: any interrupt or wake-up event.

- Stop mode

This mode puts the FLASH into low-power mode and the RC oscillator of the HSI is switched off.

Exit conditions: any external interrupt/event (EXTI signal), external reset signal on RST, IWDG reset, where EXTI signal includes one of the 24 external I/O ports, output of PVD, wake-up signal of USB, wake-up signal of USB PD, etc.

- Standby mode

This mode FLASH enters low-power mode, the RC oscillator of HSI is switched off and the system LDO enters low-power mode.

Exit conditions: any external interrupt/event (EXTI signal), external reset signal on RST, IWDG reset, where the EXTI signal includes one of the 24 external I/O ports, the output of PVD, the wake-up signal of USB, the wake-up signal of USB PD, etc.

### 1.4.8 Programmable Fast Interrupt Controller (PFIC)

The chip has a built-in Programmable Fast Interrupt Controller (PFIC) that supports up to 255 interrupt vectors, providing flexible interrupt management with minimal interrupt latency. Currently the chip manages 7 core private interrupts and 39 peripheral interrupt management, with other interrupt sources reserved. the PFIC registers are all accessible in both user and machine privileged modes.

- 2 individually maskable interrupts
- Provide one non-maskable interrupt NMI
- Support Hardware Prologue/Epilogue (HPE) without instruction overhead
- Provide 4 Vector Table Free (VTF) for faster access to interrupt service routines
- Vector table support address or instruction mode
- Interrupt nesting depth can be configured up to 2 levels

- Support interrupt tail linking

#### 1.4.9 External Interrupt/Event Controller (EXTI)

The external interrupt/event controller contains a total of 28 edge detectors for generating interrupt/event requests. Each interrupt line can be configured independently of its trigger event (rising or falling edge or double edge) and can be individually masked; a pending register maintains the status of all interrupt requests. Up to 60 general purpose I/O ports are optionally connected to 24 external interrupt lines.

#### 1.4.10 General DMA Controller

The system has a built-in general purpose DMA controller that manages 8 channels to flexibly handle high-speed data transfers between memory to memory, peripheral to memory and memory to peripheral, supporting the ring buffer approach. Each channel has dedicated hardware DMA request logic to support one or more peripheral access requests to memory, with configurable access priority, transfer length, source and destination addresses for transfers, etc.

DMA is used for the main peripherals including: General-purpose/advanced-control timers TIMx, ADC, USART, I2C, SPI.

USB and USB PD have additional dedicated independent DMA channels.

*Note: DMA and CPU access to system SRAM after arbiter arbitration.*

#### 1.4.11 Clock and Boot

The system clock source HSI is turned on by default, and the internal 48MHz RC oscillator 6 division is used as the default CPU clock when no clock is configured or after a reset. For low power modes where the clock is turned off, the system will first turn on the internal RC oscillator upon wake-up. If the clock interrupt is enabled, the software can receive the corresponding interrupt.

#### 1.4.12 Analog-to-digital Converter (ADC) and Touchkey Capacitance Detection (TKey)

The chip has an built-in 12-bit analog/digital converter (ADC) providing up to 14 external channels and 1 internal channel sample, with programmable channel sample times for single, continuous, sweep or intermittent conversion. The provision of an analogue watchdog function allows very accurate monitoring of one or more selected channels for monitoring channel signal voltages. Supports external event-triggered transitions, with trigger sources including internal signals from the on-chip timer and external pins. Supports the use of DMA operation.

ADC internal channels are internal reference supply voltage sampling channels.

Touch key capacitance detection unit, providing up to 14 detection channels, multiplexes the external channels of the ADC module. The detection results are converted to output results by the ADC module, and the touch key status is identified by the touch detection algorithm subroutine library or by user software.

*Note: The channel 3, channel 7, channel 11 and channel 15 functions of the ADC are not available for products with a lot number with the penultimate 5 digits being 0.*

#### 1.4.13 Timer and Watchdog

- Advanced-control timer (TIM1, TIM2)

The Advanced Control Timer is a 16-bit auto-loading up/down counter with a 16-bit programmable prescaler. In addition to the full general-purpose timer functionality, it can be viewed as a three-phase PWM generator assigned to 6 channels, with a complementary PWM output functionality with dead-band insertion, allowing the timer to be updated after a specified number of counter cycles for repeating Counting cycle, braking function, etc. Many functions of advanced control timers are the same as general timers, and the internal structure is also the same.



Therefore, advanced control timers can cooperate with other TIM timers through the timer link function to provide synchronization or event link functions.

- General-purpose timer (TIM3)

The general-purpose timer is a 16-bit auto-loading recursive counter with a programmable 16-bit prescaler and 2 independent channels, each supporting input capture, output comparison, PWM generation and single pulse mode output. It can also work with advanced-control timers via the timer linking function to provide synchronous or event linking functionality. In debug mode, the counter can be frozen and any general-purpose timer can be used to generate PWM outputs.

- Independent watchdog

The Independent Watchdog is a free-running 12-bit decrementing counter supporting 7 division factors. The clock source is provided by the division of the (HSI/1024) clock. IWDG works completely independently of the main program and can therefore be used to reset the entire system in case of problems or as a free timer to provide timeout management for applications. With the option byte it can be configured to be a software or hardware start watchdog. In debug mode, the counter can be frozen.

- Window watchdog

Window watchdog is a 7-bit decrementing counter and can be set to run freely. It can be used to reset the entire system in the event of a problem. It is driven by the master clock and has an early warning interrupt function; in debug mode the counter can be frozen.

- SysTick timer

QingKe microprocessor core comes with a 64-bit optional incremental or decremental counter for generating SYSTICK exceptions (exception number: 12), which can be used exclusively in real time operating systems to provide a "heartbeat" rhythm for the system, or as a standard 64-bit counter. Automatic reload function and programmable clock source.

#### 1.4.14 Universal Synchronous/Asynchronous Receiver Transmitter (USART)

The chip provides 4 sets of universal synchronous/asynchronous transceivers. It supports full duplex asynchronous serial communication, synchronous unidirectional communication as well as half duplex single line communication, also LIN (Local Interconnect Network), ISO7816 compatible smart card protocol and IrDA SIR ENDEC transmission codec specification, as well as modem (CTS/RTS hardware flow control) operation, and also supports multi-processor communication. It uses a fractional baud rate generator system and supports continuous communication by DMA operation.

#### 1.4.15 Serial Peripheral Interface (SPI)

The chip provides 1 serial peripheral SPI interface, support master or slave operation, dynamic switching. Support multi-master mode, full-duplex or half-duplex synchronous transmission, support basic SD card and MMC mode. Programmable clock polarity and phase, data bit width provides 8 or 16-bit selection, hardware CRC generation/check for reliable communication, and continuous communication support for DMA operation.

#### 1.4.16 I2C Bus

The chip provides one I2C bus interface, capable of working in multi-master or slave mode, performing all I2C bus specific timing, protocols, arbitration, etc. Both standard and fast communication speeds are supported and it is also

compatible with SMBus 2.0.

The I2C interface provides 7-bit or 10-bit addressing, and supports dual slave addressing in 7-bit Slave mode. It integrates built-in hardware CRC generator/checker. It also supports DMA operation and supports SMBus bus version 2.0/PMBus bus.

*Note: I2C function is not available for products with a 0 in the penultimate 5 digits of the lot number.*

#### **1.4.17 Universal Serial Bus USB2.0 Full-speed Host/Device Controller (USBFS)**

USB2.0 Full-speed Host Controller and Device Controller (USBFS) following the USB2.0 Full-speed standard and supporting the BC charging protocol. Provides 8 configurable USB device endpoints and a set of host endpoints. Supports control/lot/sync/interrupt transfers, double buffer mechanism, USB bus hang/resume operation and provides standby/wakeup functions. 48MHz clock dedicated to the USBFS module is generated directly from the internal high-speed clock (HSI).

#### **1.4.18 USB PD and Type-C Controller (USB PD)**

Built-in USB Power Delivery controller and PD transceiver PHY, support USB Type-C master-slave detection, automatic BMC codec and CRC, hardware edge control, support USB PD2.0 and PD3.0 power delivery control, support fast charging, support UFP/PD powered end Sink and DFP/PD powered end Source applications and DRP applications, supports PDUSB.

#### **1.4.19 General-purpose Input and Output (GPIO)**

The system provides 3 groups of GPIO ports with a total of 60 GPIO pins. Each pin can be configured by software as an output, input (with or without pull-up, some pins support pull-down) or multiplexed peripheral function port. All GPIO pins support controlled pull-up, only PA0-PA15 and PC16-PC17 support controlled pull-down, the remaining pins do not support pull-down. PC14-PC17 support multiple pull-up modes, set by dedicated control registers corresponding to the PD and USB pins respectively.

Most GPIO pins are shared with either digital or analogue multiplexed peripherals. All PA and PB GPIO pins have high current drive capability with simple constant current functionality and all support PWM. a lockout mechanism is provided to freeze the I/O configuration to avoid accidental writes to I/O registers.

Most of the I/O pins in the system are powered by  $V_{DD}$ , and changing the  $V_{DD}$  power supply will change the I/O pin output level high enough to adapt to the external communication interface level. Please refer to the pin description for specific pins.

#### **1.4.20 Operational Amplifier/Comparator (OPA)**

The chip has a built-in 2-group op-amp (OPA), which can also be used as a voltage comparator. Its input can be selected for multiple channels by changing the configuration, including amplification selection for the programmable gain op-amp (PGA), and its output can be selected for 2 channels by changing the configuration, internally associated to ADC channels. External analogue small signal amplification is supported for feeding into the ADC for small signal ADC conversion.

#### **1.4.21 Voltage Comparator (CMP)**

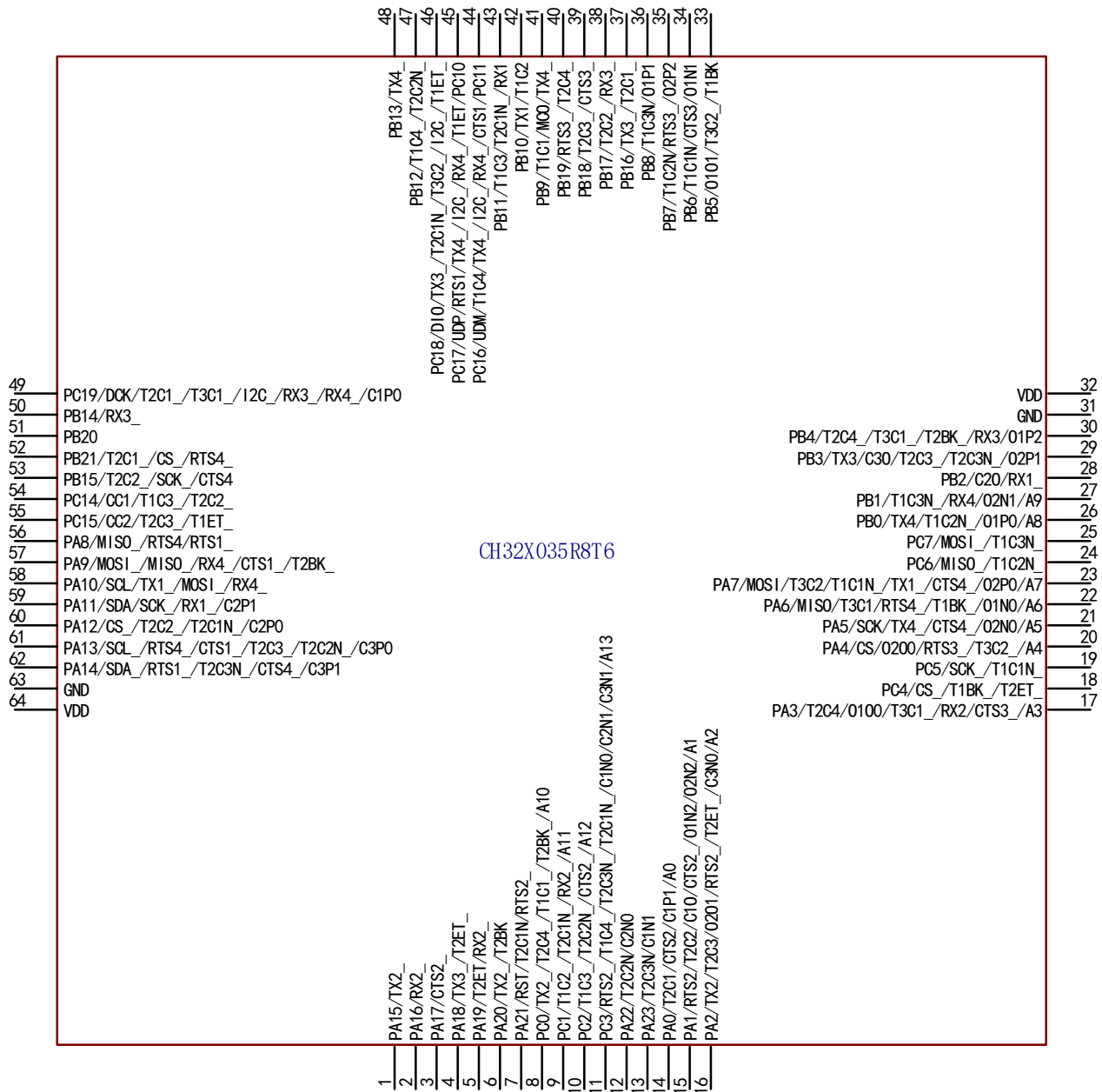
The chip has 3 built-in rail-to-rail analog voltage comparators with optional hysteresis characteristics. The voltage comparison results are triggered by the GPIO output or internally directly into the input channels CH1 ~ CH3 of the TIM2.

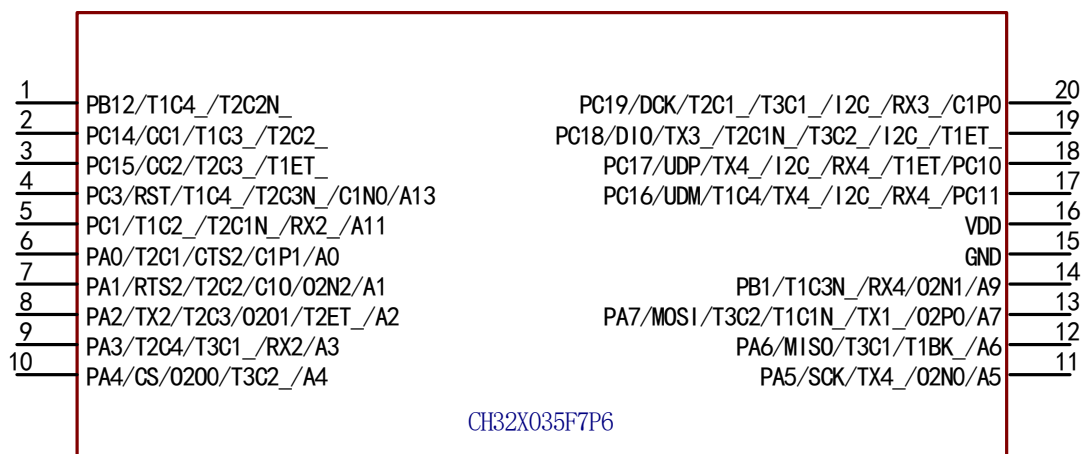
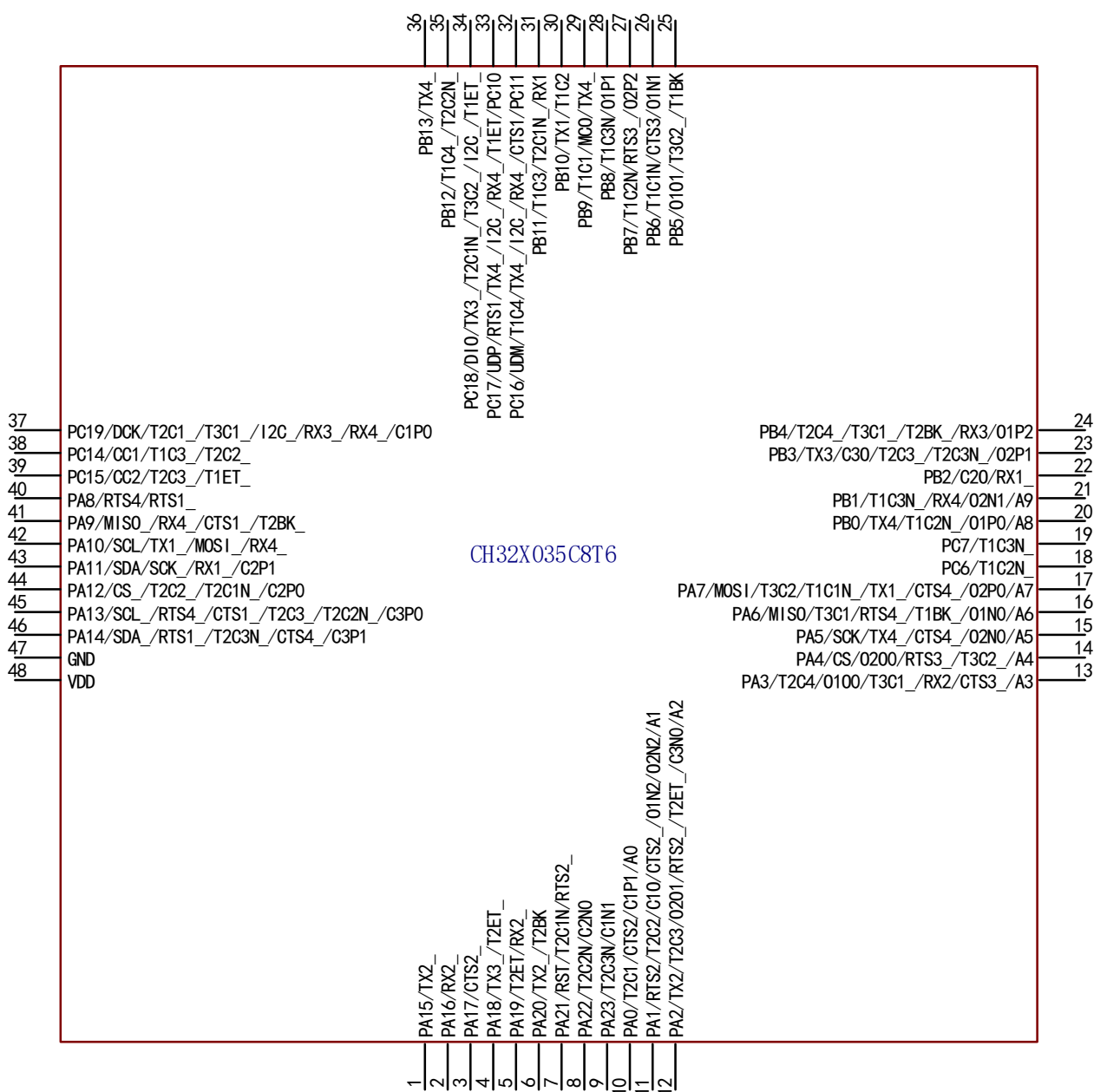
#### **1.4.22 2-wire SDI Serial Debug Interface**

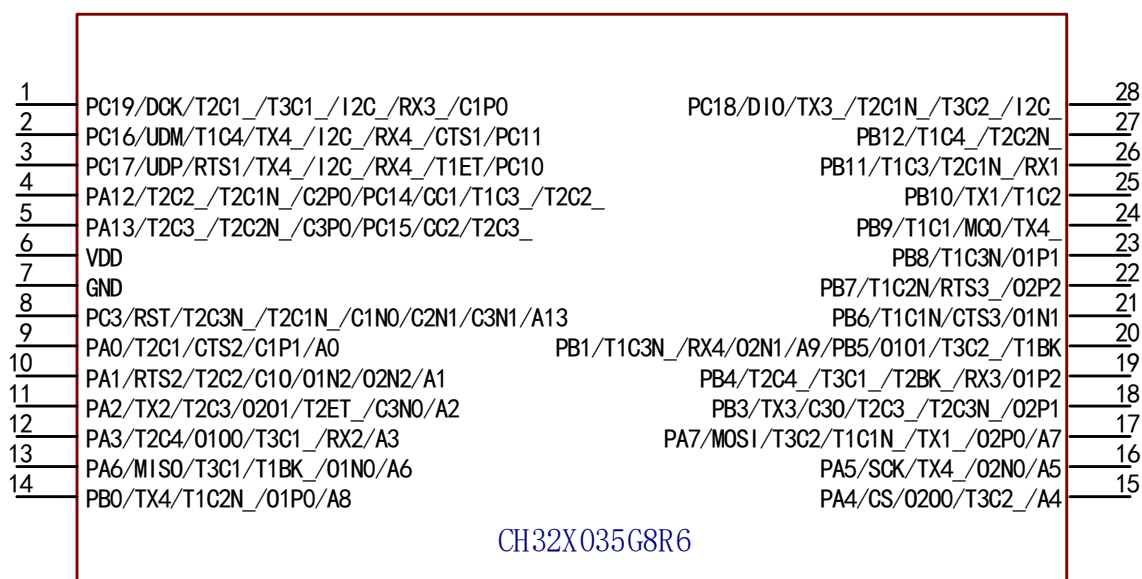
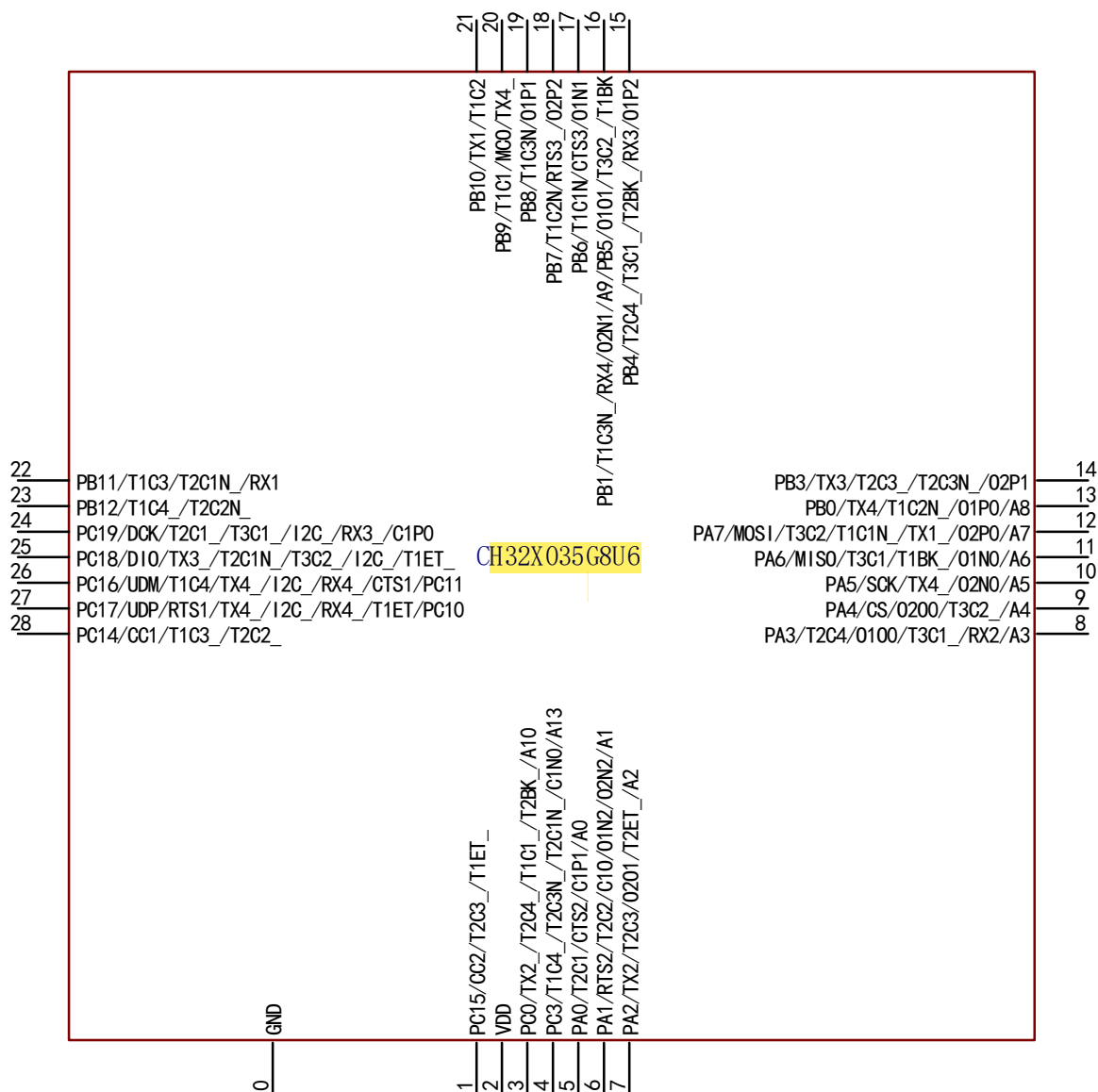
The core comes with a 2-wire serial debug interface (SDI) including the SWDIO and SWCLK pins. The default debug interface pin function is turned on after system power-up or reset, and the SDI can be turned off as required after the main program is running.

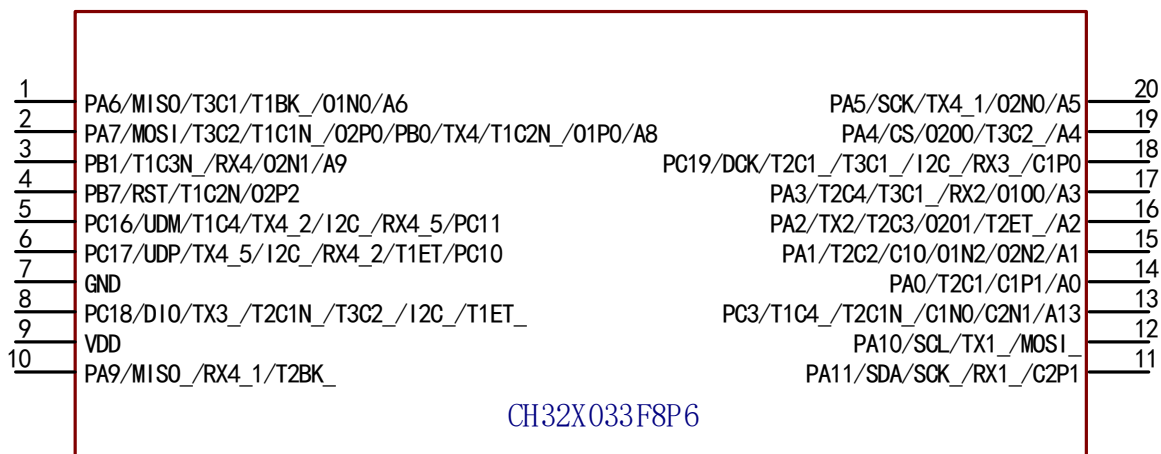
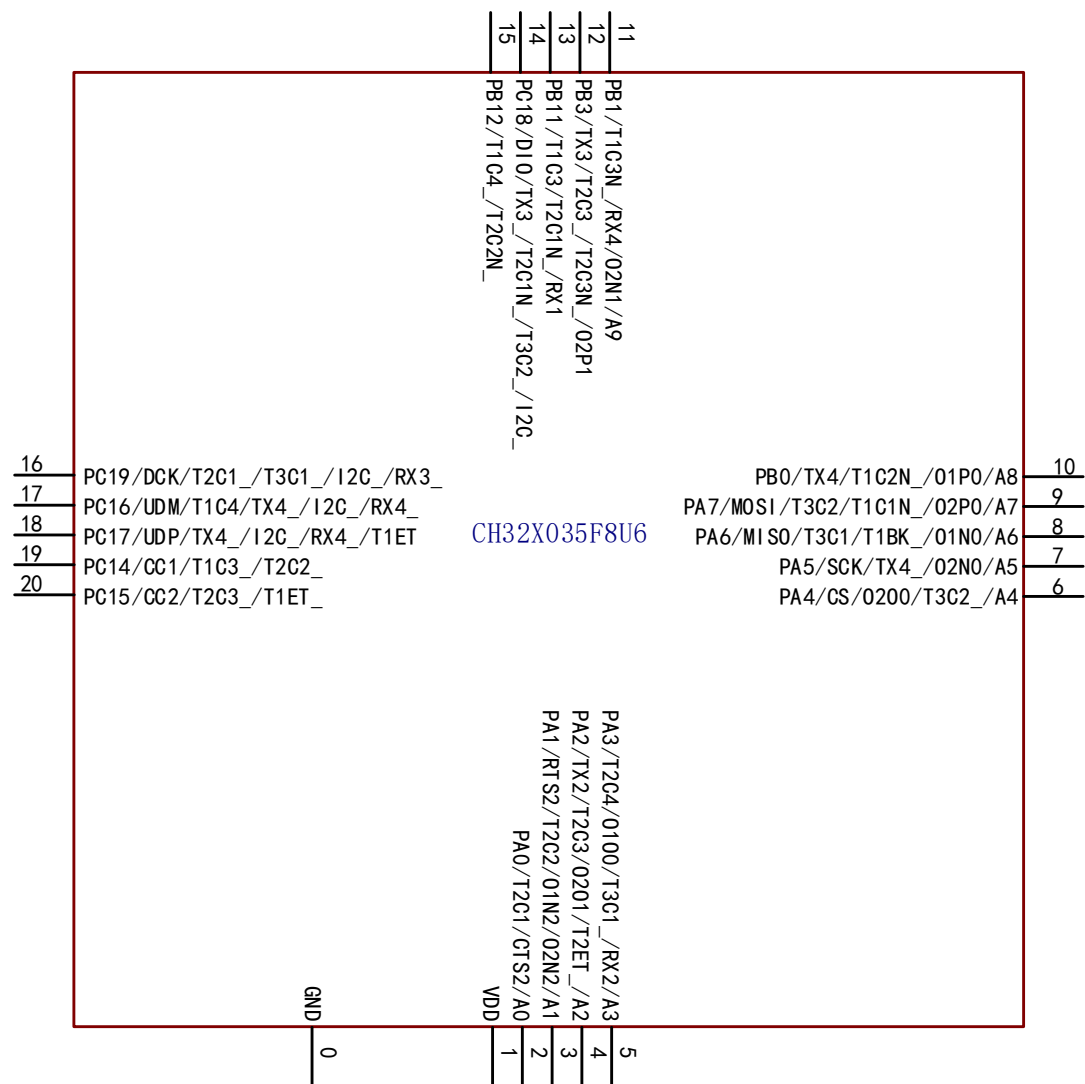
## Chapter 2 Pinouts and Pin Definition

### 2.1 Pinouts









Note: The multiplexed functions in the pin diagram are abbreviated.

Example: A:ADC\_ (A10:ADC\_IN10)

C:COMP\_ (C3N0:COMP3\_N0)

T:TIME\_ (T2C4:TIM2\_CH4、T2C2N:TIM2\_CH2N)

O:OPA\_ (O1N2:OPA1\_N2、 O2O0:OPA2\_OUT0)  
TX2:USART2\_TX  
CS:SPI\_NSS  
UDP:USBDP  
UDM:USBDM  
DIO:SWDIO  
DCK:SWCLK



## 2.2 Pin Description

Note that the pin function descriptions in the table below are for all functions and do not relate to specific product models. Peripheral resources may vary between models, so please check the availability of this feature against the product model resource table before viewing.

Table 2-1 CH32X035 Pin definitions

| Pin No. |        |       |        |       |         | Pin name        | Pin type <sup>(1)</sup> | Main function (after reset) | Default alternate function       | Remapping function                       |
|---------|--------|-------|--------|-------|---------|-----------------|-------------------------|-----------------------------|----------------------------------|------------------------------------------|
| LQFP64M | LQFP48 | QFN28 | QSOP28 | QFN20 | TSSOP20 |                 |                         |                             |                                  |                                          |
| -       | -      | 0     | -      | 0     | -       | GND             | P                       | GND                         |                                  |                                          |
| 31      | 47     | -     | 7      | -     | 15      | GND             | P                       | GND                         |                                  |                                          |
| 1       | 1      | -     | -      | -     | -       | PA15            | I/O                     | PA15                        |                                  | TX2_2/TX2_4                              |
| 2       | 2      | -     | -      | -     | -       | PA16            | I/O                     | PA16                        |                                  | RX2_2/RX2_4                              |
| 3       | 3      | -     | -      | -     | -       | PA17            | I/O                     | PA17                        |                                  | CTS2_2/CTS2_4                            |
| 4       | 4      | -     | -      | -     | -       | PA18            | I/O                     | PA18                        |                                  | TX3_2/T2ET_1/T2ET_3                      |
| 5       | 5      | -     | -      | -     | -       | PA19            | I/O                     | PA19                        | T2ET                             | RX2_1/T2ET_2                             |
| 6       | 6      | -     | -      | -     | -       | PA20            | I/O                     | PA20                        | T2BK                             | TX2_1/T2BK_2                             |
| 7       | 7      | -     | -      | -     | -       | PA21            | I/O                     | PA21                        | RST/T2C1N                        | RTS2_2                                   |
| 8       | -      | 3     | -      | -     | -       | PC0             | I/O/A                   | PC0                         | A10                              | TX2_3/T2C4_5<br>/T2C4_6/T1C1_3/T2BK_4    |
| 9       | -      | -     | -      | -     | 5       | PC1             | I/O/A                   | PC1                         | A11 <sup>(3)</sup>               | T1C2_3/T2C1N_4/RX2_3                     |
| 10      | -      | -     | -      | -     | -       | PC2             | I/O/A                   | PC2                         | A12                              | T1C3_3/T2C2N_4/CTS2_3                    |
| 11      | -      | 4     | -      | -     | -       | PC3             | I/O/A                   | PC3                         | C1N0/C2N1/C3N1/A13               | RTS2_3/T1C4_3<br>/T2C3N_4/T2C1N_2/RTS2_4 |
| -       | -      | -     | 8      | -     | 4       |                 |                         |                             | RST/C1N0/C2N1<br>/C3N1/A13       |                                          |
| 12      | 8      | -     | -      | -     | -       | PA22            | I/O/A                   | PA22                        | T2C2N/C2N0                       | CK2_2/T2C2N_2/CK2_4                      |
| 13      | 9      | -     | -      | -     | -       | PA23            | I/O/A                   | PA23                        | T2C3N/C1N1                       | CK2_1/T2C3N_2                            |
| 14      | 10     | 5     | 9      | 2     | 6       | PA0             | I/O/A                   | PA0                         | T2C1/CTS2/C1P1/A0                | T2C1_2                                   |
| 15      | 11     | 6     | 10     | 3     | 7       | PA1             | I/O/A                   | PA1                         | RTS2/T2C2/C1O<br>/O1N2/O2N2/A1   | CTS2_1/T2C2_2                            |
| 16      | 12     | 7     | 11     | 4     | 8       | PA2             | I/O/A                   | PA2                         | TX2/T2C3/O2O1/C3N0<br>/A2        | RTS2_1/T2ET_5/T2C3_1/T2ET_6              |
| 63      | -      | -     | -      | -     | -       | GND             | P                       | GND                         |                                  |                                          |
| 32      | 48     | 2     | 6      | 1     | 16      | V <sub>DD</sub> | P                       | VDD                         |                                  |                                          |
| 17      | 13     | 8     | 12     | 5     | 9       | PA3             | I/O/A                   | PA3                         | RX2/T2C4/O1O0/A3 <sup>(3)</sup>  | T3C1_3/T2C4_1/CTS3_2                     |
| 18      | -      | -     | -      | -     | -       | PC4             | I/O                     | PC4                         |                                  | CS_3/T1BK_3/T2ET_4                       |
| 19      | -      | -     | -      | -     | -       | PC5             | I/O                     | PC5                         |                                  | SCK_3/T1C1N_3                            |
| 20      | 14     | 9     | 15     | 6     | 10      | PA4             | I/O/A                   | PA4                         | CS/CK2/O2O0/A4                   | RTS3_2/T3C2_3                            |
| 21      | 15     | 10    | 16     | 7     | 11      | PA5             | I/O/A                   | PA5                         | SCK/O2N0/A5                      | TX4_1/CTS4_4                             |
| 22      | 16     | 11    | 13     | 8     | 12      | PA6             | I/O/A                   | PA6                         | MISO/T3C1/O1N0/A6                | CK4_1/RTS4_4/T1BK_1                      |
| 23      | 17     | 12    | 17     | 9     | 13      | PA7             | I/O/A                   | PA7                         | MOSI/T3C2/O2P0/A7 <sup>(3)</sup> | T1C1N_1/TX1_3/CTS4_1                     |

| Pin No. |        |       |        |       |         | Pin name            | Pin type <sup>(1)</sup> | Main function (after reset) | Default alternate function | Remapping function                                                               |
|---------|--------|-------|--------|-------|---------|---------------------|-------------------------|-----------------------------|----------------------------|----------------------------------------------------------------------------------|
| LQFP64M | LQFP48 | QFN28 | QSOP28 | QFN20 | TSSOP20 |                     |                         |                             |                            |                                                                                  |
| 24      | 18     | -     | -      | -     | -       | PC6                 | I/O                     | PC6                         |                            | MISO_3/T1C2N_3                                                                   |
| 25      | 19     | -     | -      | -     | -       | PC7                 | I/O                     | PC7                         |                            | MOSI_3/T1C3N_3/PIOC_IO0_1                                                        |
| 26      | 20     | 13    | 14     | 10    | -       | PB0                 | I/O/A                   | PB0                         | TX4/O1P0/A8                | T1C2N_1                                                                          |
| 27      | 21     | 16    | 20     | 11    | 14      | PB1 <sup>(5)</sup>  | I/O/A                   | PB1                         | RX4/O2N1/A9                | T1C3N_1                                                                          |
| 28      | 22     | -     | -      | -     | -       | PB2                 | I/O/A                   | PB2                         | CK4/C2O                    | RX1_3/CK4_2/CK4_5                                                                |
| 29      | 23     | 14    | 18     | 12    | -       | PB3                 | I/O/A                   | PB3                         | TX3/C3O/O2P1               | T2C3_2/T2C3N_5/T2C3_3/T2C3N_6                                                    |
| 30      | 24     | 15    | 19     | -     | -       | PB4                 | I/O/A                   | PB4                         | RX3/O1P2                   | T2C4_2/T3C1_1/T2BK_5<br>/T2C4_3/T2BK_6                                           |
| 64      | -      | -     | -      | -     | -       | V <sub>DD</sub>     | P                       | V <sub>DD</sub>             |                            |                                                                                  |
| 33      | 25     | 16    | 20     | -     | -       | PB5 <sup>(5)</sup>  | I/O/A                   | PB5                         | CK3/O1O1/T1BK              | CK1_2/T3C2_1/CK3_1/T1BK_2                                                        |
| 34      | 26     | 17    | 21     | -     | -       | PB6                 | I/O/A                   | PB6                         | T1C1N/CTS3/O1N1            | T1C1N_2/CTS3_1                                                                   |
| 35      | 27     | 18    | 22     | -     | -       | PB7                 | I/O/A                   | PB7                         | T1C2N/O2P2/RTS3            | RTS3_1/T1C2N_2                                                                   |
| 36      | 28     | 19    | 23     | -     | -       | PB8                 | I/O/A                   | PB8                         | T1C3N/O1P1                 | CK3_2/CK4_3/T1C3N_2                                                              |
| 37      | -      | -     | -      | -     | -       | PB16                | I/O                     | PB16                        |                            | TX3_3/T2C1_4                                                                     |
| 38      | -      | -     | -      | -     | -       | PB17                | I/O                     | PB17                        |                            | T2C2_4/RX3_3                                                                     |
| 39      | -      | -     | -      | -     | -       | PB18                | I/O                     | PB18                        |                            | T2C3_4/CTS3_3                                                                    |
| 40      | -      | -     | -      | -     | -       | PB19                | I/O                     | PB19                        |                            | RTS3_3/T2C4_4                                                                    |
| 41      | 29     | 20    | 24     | -     | -       | PB9                 | I/O                     | PB9                         | CK1/T1C1/MCO               | TX4_3/CK1_1/T1C1_1/T1C1_2                                                        |
| 42      | 30     | 21    | 25     | -     | -       | PB10                | I/O                     | PB10                        | TX1/T1C2                   | T1C2_1/T1C2_2/TX1_2                                                              |
| 43      | 31     | 22    | 26     | 13    | -       | PB11                | I/O                     | PB11                        | T1C3/RX1                   | T1C3_1/T1C3_2/RX1_2/T2C1N_6                                                      |
| 44      | 32     | 26    | 2      | 17    | 17      | PC16 <sup>(4)</sup> | I/O/A                   | PC16                        | UDM/T1C4/CTS1              | TX4_2/SCL_2 <sup>(3)</sup> /SDA_4 <sup>(3)</sup> /RX4_5<br>/CTS1_1/T1C4_1        |
|         |        |       |        | -     |         | PC11 <sup>(4)</sup> | I/O                     | PC11                        |                            |                                                                                  |
| 45      | 33     | 27    | 3      | 18    | 18      | PC17 <sup>(4)</sup> | I/O/A                   | PC17                        | UDP/RTS1/T1ET              | TX4_5/SDA_2 <sup>(3)</sup> /SCL_4 <sup>(3)</sup> /RX4_2<br>/RTS1_1/T1ET_1        |
|         |        |       |        | -     |         | PC10 <sup>(4)</sup> | I/O                     | PC10                        |                            |                                                                                  |
| 46      | 34     | 25    | 28     | 14    | 19      | PC18                | I/O                     | PC18                        | DIO/PIOC_IO0               | TX3_1/T2C1N_5/SDA_3 <sup>(3)</sup> /SCL_5 <sup>(3)</sup><br>T1ET_2/T1ET_3/T3C2_2 |
| 47      | 35     | 23    | 27     | 15    | 1       | PB12                | I/O                     | PB12                        |                            | CK1_3/T1C4_2/T2C2N_5/T2C2N_6                                                     |
| 48      | 36     | -     | -      | -     | -       | PB13                | I/O                     | PB13                        |                            | TX4_4                                                                            |
| 49      | 37     | 24    | 1      | 16    | 20      | PC19                | I/O/A                   | PC19                        | DCK/PIOC_IO1/C1P0          | T2C1_5/T3C1_2/SCL_3 <sup>(3)</sup> /SDA_5 <sup>(3)</sup><br>/RX3_1/RX4_4/T2C1_6  |
| 50      | -      | -     | -      | -     | -       | PB14                | I/O                     | PB14                        |                            | RX3_2                                                                            |
| 51      | -      | -     | -      | -     | -       | PB20                | I/O                     | PB20                        |                            | CK2_3                                                                            |
| 52      | -      | -     | -      | -     | -       | PB21                | I/O                     | PB21                        |                            | T2C1_1/CS_1/RTS4_1/T2C1_3                                                        |
| 53      | -      | -     | -      | -     | -       | PB15                | I/O                     | PB15                        | CTS4                       | T2C2_1/SCK_1/T2C2_3/CTS4_2<br>/CTS4_5                                            |

| Pin No. |        |       |        |       |         | Pin name            | Pin type <sup>(1)</sup> | Main function (after reset) | Default alternate function | Remapping function                                         |
|---------|--------|-------|--------|-------|---------|---------------------|-------------------------|-----------------------------|----------------------------|------------------------------------------------------------|
| LQFP64M | LQFP48 | QFN28 | QSOP28 | QFN20 | TSSOP20 |                     |                         |                             |                            |                                                            |
| 54      | 38     | 28    | 4      | 19    | 2       | PC14                | I/O/A                   | PC14                        | CC1                        | T1C3_4/T2C2_6                                              |
| 55      | 39     | 1     | 5      | 20    | 3       | PC15                | I/O/A                   | PC15                        | CC2                        | T2C3_6/T1ET_4                                              |
| 56      | 40     | -     | -      | -     | -       | PA8                 | I/O                     | PA8                         | RTS4                       | RTS1_2/CK4_4/RTS4_2/RTS4_5/MISO_1                          |
| 57      | 41     | -     | -      | -     | -       | PA9                 | I/O                     | PA9                         |                            | MOSI_1/RX4_1/CTS1_2/MISO_2/T2BK_1/T2BK_3                   |
| 58      | 42     | -     | -      | -     | -       | PA10                | I/O                     | PA10                        | SCL <sup>(3)</sup>         | TX1_1/MOSI_2/RX4_3                                         |
| 59      | 43     | -     | -      | -     | -       | PA11                | I/O/A                   | PA11                        | SDA <sup>(3)</sup> /C2P1   | SCK_2/RX1_1                                                |
| 60      | 44     | -     | 4      | -     | -       | PA12 <sup>(6)</sup> | I/O/A                   | PA12                        | C2P0                       | CS_2/T2C2_5/T2C1N_1/T2C1N_3                                |
| 61      | 45     | -     | 5      | -     | -       | PA13 <sup>(6)</sup> | I/O/A                   | PA13                        | C3P0                       | SCL_1 <sup>(3)</sup> /RTS4_3/CTS1_3/T2C3_5/T2C2N_1/T2C2N_3 |
| 62      | 46     | -     | -      | -     | -       | PA14                | I/O/A                   | PA14                        | C3P1                       | SDA_1 <sup>(3)</sup> /RTS1_3/T2C3N_1/CTS4_3/T2C3N_3        |

Table 2-2 CH32X033 Pin definitions

| Pin No. |         | Pin name           | Pin type <sup>(1)</sup> | Main function (after reset) | Default alternate function       | Remapping function                       |
|---------|---------|--------------------|-------------------------|-----------------------------|----------------------------------|------------------------------------------|
|         | TSSOP20 |                    |                         |                             |                                  |                                          |
| -       | 7       | GND                | P                       | GND                         |                                  |                                          |
| -       | 9       | V <sub>DD</sub>    | P                       | VDD                         |                                  |                                          |
| -       | 14      | PA0                | I/O/A                   | PA0                         | T2C1/CTS2/C1P1/A0                | T2C1_2                                   |
| -       | 15      | PA1                | I/O/A                   | PA1                         | RTS2/T2C2/C1O/O1N2/O2N2/A1       | CTS2_1/T2C2_2                            |
| -       | 16      | PA2                | I/O/A                   | PA2                         | TX2/T2C3/O2O1/C3N0/A2            | RTS2_1/T2ET_5/T2C3_1/T2ET_6              |
| -       | 17      | PA3                | I/O/A                   | PA3                         | RX2/T2C4/O1O0/A3 <sup>(3)</sup>  | T3C1_3/T2C4_1/CTS3_2                     |
| -       | 19      | PA4                | I/O/A                   | PA4                         | CS/CK2/O2O0/A4                   | RTS3_2/T3C2_3                            |
| -       | 20      | PA5                | I/O/A                   | PA5                         | SCK/O2N0/A5                      | TX4_1/CTS4_4                             |
| -       | 1       | PA6                | I/O/A                   | PA6                         | MISO/T3C1/O1N0/A6                | CK4_1/RTS4_4/T1BK_1                      |
| -       | 2       | PA7 <sup>(7)</sup> | I/O/A                   | PA7                         | MOSI/T3C2/O2P0/A7 <sup>(3)</sup> | T1C1N_1/TX1_3/CTS4_1                     |
| -       | 10      | PA9                | I/O                     | PA9                         |                                  | MOSI_1/RX4_1/CTS1_2/MISO_2/T2BK_1/T2BK_3 |
| -       | 12      | PA10               | I/O                     | PA10                        | SCL <sup>(3)</sup>               | TX1_1/MOSI_2/RX4_3                       |
| -       | 11      | PA11               | I/O/A                   | PA11                        | SDA <sup>(3)</sup> /C2P1         | SCK_2/RX1_1                              |
| -       | 2       | PB0 <sup>(7)</sup> | I/O/A                   | PB0                         | TX4/O1P0/A8                      | T1C2N_1                                  |
| -       | 3       | PB1                | I/O/A                   | PB1                         | RX4/O2N1/A9                      | T1C3N_1                                  |
| -       | 4       | PB7                | I/O/A                   | PB7                         | RST/T1C2N/O2P2/RTS3              | RTS3_1/T1C2N_2                           |

| Pin No. |         | Pin name            | Pin type <sup>(1)</sup> | Main function (after reset) | Default alternate function | Remapping function                                                               |
|---------|---------|---------------------|-------------------------|-----------------------------|----------------------------|----------------------------------------------------------------------------------|
|         | TSSOP20 |                     |                         |                             |                            |                                                                                  |
| -       | 13      | PC3                 | I/O/A                   | PC3                         | C1N0/C2N1/C3N1/A13         | RTS2_3/T1C4_3<br>/T2C3N_4/T2C1N_2/RTS2_4                                         |
| -       | 5       | PC16 <sup>(4)</sup> | I/O/A                   | PC16                        | UDM/T1C4/CTS1              | TX4_2/SCL_2 <sup>(3)</sup> /SDA_4 <sup>(3)</sup> /RX4_5<br>/CTS1_1/T1C4_1        |
|         |         | PC11 <sup>(4)</sup> | I/O                     | PC11                        |                            |                                                                                  |
| -       | 6       | PC17 <sup>(4)</sup> | I/O/A                   | PC17                        | UDP/RTS1/T1ET              | TX4_5/SDA_2 <sup>(3)</sup> /SCL_4 <sup>(3)</sup> /RX4_2<br>/RTS1_1/T1ET_1        |
|         |         | PC10 <sup>(4)</sup> | I/O                     | PC10                        |                            |                                                                                  |
| -       | 8       | PC18                | I/O                     | PC18                        | DIO/PIOC_IO0               | TX3_1/T2C1N_5/SDA_3 <sup>(3)</sup> /SCL_5 <sup>(3)</sup><br>T1ET_2/T1ET_3/T3C2_2 |
| -       | 18      | PC19                | I/O/A                   | PC19                        | DCK/PIOC_IO1/C1P0          | T2C1_5/T3C1_2/SCL_3 <sup>(3)</sup> /SDA_5 <sup>(3)</sup><br>/RX3_1/RX4_4/T2C1_6  |

Note 1: Explanation of table abbreviations:

I = TTL/CMOS level Schmitt input; O = CMOS level tri-state output;

A = analog signal input or output; P = power supply.

Note 2: The value after the remapping function underline indicates the configuration value of the corresponding bit in the AFIO register. For example: TX2\_2 indicates that the corresponding bit of the AFIO register is configured as 10b.

Note 3: The channel 3, channel 7, channel 11, channel 15 and I2C functions of the ADC are not applicable to products with a lot number with the penultimate bit 5 being 0.

Note 4: Except for the CH32X035F8U6 chip (QFN20 package), for CH32X033 and other CH32X035 model chips, the PC10 and PC17 pins are short-joined inside the chip, and both IOs are prohibited from being configured as output functions; the PC11 and PC16 pins are short-joined inside the chip. Seal, prohibiting both IOs from being configured as output functions; in USB applications, the PC10 and PC11 pins should be configured in floating input mode (default value after reset).

Note 5: For CH32X035G8U6 and CH32X035G8R6 chips, PB1 and PB5 pins are shorted and sealed inside the chip, prohibiting both IOs from being configured for output function.

Note 6: For CH32X035G8R6 chip, PA12 and PC14 pins are shorted and sealed inside the chip, prohibiting both IOs from being configured for output function; PA13 and PC15 pins are shorted and sealed inside the chip, prohibiting both IOs from being configured for output function.

Note 7: PA7 and PB0 pins are short-circuited inside the chip, prohibiting both IOs from being configured as output functions.

## 2.3 Pin Alternate Functions

Note: The pin function in the table below refer to all functions and does not involve specific model(s). There are differences in peripheral resources between different models. Please confirm whether this function is available according to the particular model's resource table before viewing this table.

Table 2-3 Pin alternate and remapping functions

| Alternate Pin | ADC               | TIM1/2                             | TIM3   | USART                                       | CMP  | SYS | I2C                  | SPI              | USB | OPA          | PIOC |
|---------------|-------------------|------------------------------------|--------|---------------------------------------------|------|-----|----------------------|------------------|-----|--------------|------|
| PA0           | A0                | T2C1<br>T2C1_2                     |        | CTS2                                        | C1P1 |     |                      |                  |     |              |      |
| PA1           | A1                | T2C2<br>T2C2_2                     |        | RTS2<br>CTS2_1                              | C1O  |     |                      |                  |     | O1N2<br>O2N2 |      |
| PA2           | A2                | T2C3<br>T2C3_1<br>T2ET_5<br>T2ET_6 |        | TX2<br>RTS2_1                               | C3N0 |     |                      |                  |     | O2O1         |      |
| PA3           | A3 <sup>(1)</sup> | T2C4<br>T2C4_1                     | T3C1_3 | RX2<br>CTS3_2                               |      |     |                      |                  |     | O1O0         |      |
| PA4           | A4                |                                    | T3C2_3 | CK2<br>RTS3_2                               |      |     |                      | CS               |     | O2O0         |      |
| PA5           | A5                |                                    |        | TX4_1<br>CTS4_4                             |      |     |                      | SCK              |     | O2N0         |      |
| PA6           | A6                | T1BK_1                             | T3C1   | CK4_1<br>RTS4_4                             |      |     |                      | MISO             |     | O1N0         |      |
| PA7           | A7 <sup>(1)</sup> | T1C1N_1                            | T3C2   | CTS4_1<br>TX1_3                             |      |     |                      | MOSI             |     | O2P0         |      |
| PA8           |                   |                                    |        | RTS4<br>RTS1_2<br>CK4_4<br>RTS4_2<br>RTS4_5 |      |     |                      | MISO_1           |     |              |      |
| PA9           |                   | T2BK_1<br>T2BK_3                   |        | RX4_1<br>CTS1_2                             |      |     |                      | MOSI_1<br>MISO_2 |     |              |      |
| PA10          |                   |                                    |        | TX1_1<br>RX4_3                              |      |     | SCL <sup>(1)</sup>   | MOSI_2           |     |              |      |
| PA11          |                   |                                    |        | RX1_1                                       | C2P1 |     | SDA <sup>(1)</sup>   | SCK_2            |     |              |      |
| PA12          |                   | T2C2_5<br>T2C1N_1<br>T2C1N_3       |        |                                             | C2P0 |     |                      | CS_2             |     |              |      |
| PA13          |                   | T2C3_5<br>T2C2N_1<br>T2C2N_3       |        | RTS4_3<br>CTS1_3                            | C3P0 |     | SCL_1 <sup>(1)</sup> |                  |     |              |      |
| PA14          |                   | T2C3N_1<br>T2C3N_3                 |        | CTS4_3<br>RTS1_3                            | C3P1 |     | SDA_1 <sup>(1)</sup> |                  |     |              |      |
| PA15          |                   |                                    |        | TX2_2<br>TX2_4                              |      |     |                      |                  |     |              |      |
| PA16          |                   |                                    |        | RX2_2<br>RX2_4                              |      |     |                      |                  |     |              |      |
| PA17          |                   |                                    |        | CTS2_2<br>CTS2_4                            |      |     |                      |                  |     |              |      |
| PA18          |                   | T2ET_1<br>T2ET_3                   |        | TX3_2                                       |      |     |                      |                  |     |              |      |
| PA19          |                   | T2ET<br>T2ET_2                     |        | RX2_1                                       |      |     |                      |                  |     |              |      |
| PA20          |                   | T2BK<br>T2BK_2                     |        | TX2_1                                       |      |     |                      |                  |     |              |      |
| PA21          |                   | T2C1N                              |        | RTS2_2                                      |      | RST |                      |                  |     |              |      |
| PA22          |                   | T2C2N<br>T2C2N_2                   |        | CK2_2<br>CK2_4                              | C2NO |     |                      |                  |     |              |      |
| PA23          |                   | T2C3N<br>T2C3N_2                   |        | CK2_1                                       | C1N1 |     |                      |                  |     |              |      |
| PB0           | A8                | T1C2N_1                            |        | TX4                                         |      |     |                      |                  |     | O1P0         |      |
| PB1           | A9                | T1C3N_1                            |        | RX4                                         |      |     |                      |                  |     | O2N1         |      |
| PB2           |                   |                                    |        | RX1_3<br>CK4<br>CK4_2<br>CK4_5              | C2O  |     |                      |                  |     |              |      |

| Alternate<br>Pin | ADC                | TIM1/2                                 | TIM3   | USART                            | CMP                  | SYS                | I2C                                          | SPI    | USB | OPA  | PIOC       |
|------------------|--------------------|----------------------------------------|--------|----------------------------------|----------------------|--------------------|----------------------------------------------|--------|-----|------|------------|
| PB3              |                    | T2C3_2<br>T2C3_3<br>T2C3N_5<br>T2C3N_6 |        | TX3                              | C3O                  |                    |                                              |        |     | O2P1 |            |
| PB4              |                    | T2C4_2<br>T2C4_3<br>T2BK_5<br>T2BK_6   | T3C1_1 | RX3                              |                      |                    |                                              |        |     | O1P2 |            |
| PB5              |                    | T1BK<br>T1BK_2                         | T3C2_1 | CK3<br>CK1_2<br>CK3_1            |                      |                    |                                              |        |     | O1O1 |            |
| PB6              |                    | T1C1N<br>T1C1N_2                       |        | CTS3<br>CTS3_1                   |                      |                    |                                              |        |     | O1N1 |            |
| PB7              |                    | T1C2N<br>T1C2N_2                       |        | RTS3<br>RTS3_1                   |                      | RST <sup>(3)</sup> |                                              |        |     | O2P2 |            |
| PB8              |                    | T1C3N<br>T1C3N_2                       |        | CK3_2<br>CK4_3                   |                      |                    |                                              |        |     | O1P1 |            |
| PB9              |                    | T1C1<br>T1C1_1<br>T1C1_2               |        | CK1<br>CK1_1<br>TX4_3            |                      | MCO                |                                              |        |     |      |            |
| PB10             |                    | T1C2<br>T1C2_1<br>T1C2_2               |        | TX1<br>TX1_2                     |                      |                    |                                              |        |     |      |            |
| PB11             |                    | T1C3<br>T1C3_1<br>T1C3_2<br>T2C1N_6    |        | RX1<br>RX1_2                     |                      |                    |                                              |        |     |      |            |
| PB12             |                    | T1C4_2<br>T2C2N_5<br>T2C2N_6           |        | CK1_3                            |                      |                    |                                              |        |     |      |            |
| PB13             |                    |                                        |        | TX4_4                            |                      |                    |                                              |        |     |      |            |
| PB14             |                    |                                        |        | RX3_2                            |                      |                    |                                              |        |     |      |            |
| PB15             |                    | T2C2_1<br>T2C2_3                       |        | CTS4<br>CTS4_2<br>CTS4_5         |                      |                    |                                              | SCK_1  |     |      |            |
| PB16             |                    | T2C1_4                                 |        | TX3_3                            |                      |                    |                                              |        |     |      |            |
| PB17             |                    | T2C2_4                                 |        | RX3_3                            |                      |                    |                                              |        |     |      |            |
| PB18             |                    | T2C3_4                                 |        | CTS3_3                           |                      |                    |                                              |        |     |      |            |
| PB19             |                    | T2C4_4                                 |        | RTS3_3                           |                      |                    |                                              |        |     |      |            |
| PB20             |                    |                                        |        | CK2_3                            |                      |                    |                                              |        |     |      |            |
| PB21             |                    | T2C1_1<br>T2C1_3                       |        | RTS4_1                           |                      |                    |                                              | CS_1   |     |      |            |
| PC0              | A10                | T1C1_3<br>T2C4_5<br>T2C4_6<br>T2BK_4   |        | TX2_3                            |                      |                    |                                              |        |     |      |            |
| PC1              | A11 <sup>(1)</sup> | T1C2_3<br>T2C1N_4                      |        | RX2_3                            |                      |                    |                                              |        |     |      |            |
| PC2              | A12                | T1C3_3<br>T2C2N_4                      |        | CTS2_3                           |                      |                    |                                              |        |     |      |            |
| PC3              | A13                | T1C4_3<br>T2C3N_4<br>T2C1N_2           |        | RTS2_3<br>RTS2_4                 | C1NO<br>C2N1<br>C3N1 | RST <sup>(2)</sup> |                                              |        |     |      |            |
| PC4              |                    | T1BK_3<br>T2ET_4                       |        |                                  |                      |                    |                                              | CS_3   |     |      |            |
| PC5              |                    | T1C1N_3                                |        |                                  |                      |                    |                                              | SCK_3  |     |      |            |
| PC6              |                    | T1C2N_3                                |        |                                  |                      |                    |                                              | MISO_3 |     |      |            |
| PC7              |                    | T1C3N_3                                |        |                                  |                      |                    |                                              | MOSI_3 |     |      | PIOC_IO0_1 |
| PC14             |                    | T1C3_4<br>T2C2_6                       |        |                                  |                      |                    |                                              |        | CC1 |      |            |
| PC15             |                    | T1ET_4<br>T2C3_6                       |        |                                  |                      |                    |                                              |        | CC2 |      |            |
| PC16             |                    | T1C4<br>T1C4_1                         |        | CTS1<br>CTS1_1<br>TX4_2<br>RX4_5 |                      |                    | SCL_2 <sup>(1)</sup><br>SDA_4 <sup>(1)</sup> |        | UDM |      |            |
| PC17             |                    | T1ET                                   |        | RTS1                             |                      |                    | SDA_2 <sup>(1)</sup>                         |        | UDP |      |            |

| Alternate<br>Pin | ADC | TIM1/2                      | TIM3   | USART                    | CMP  | SYS | I2C                                          | SPI | USB | OPA | PIOC     |
|------------------|-----|-----------------------------|--------|--------------------------|------|-----|----------------------------------------------|-----|-----|-----|----------|
|                  |     | TIET_1                      |        | RTS1_1<br>RX4_2<br>TX4_5 |      |     | SCL_4 <sup>(1)</sup>                         |     |     |     |          |
| PC18             |     | TIET_2<br>TIET_3<br>T2C1N_5 | T3C2_2 | TX3_1                    |      | DIO | SDA_3 <sup>(1)</sup><br>SCL_5 <sup>(1)</sup> |     |     |     | PIOC_IO0 |
| PC19             |     | T2C1_5<br>T2C1_6            | T3C1_2 | RX3_1<br>RX4_4           | C1P0 | DCK | SCL_3 <sup>(1)</sup><br>SDA_5 <sup>(1)</sup> |     |     |     | PIOC_IO1 |

Note: 1. Channel 3, channel 7, channel 11, channel 15 and I2C functions of the ADC are not available for products with a lot number with a penultimate 5 digit of 0;

2. The RST function on pin PC3 is only available for the CH32X035 in the QSOP28 package and TSSOP20 package.

3. The RST function on pin PB7 is only available for CH32X033 in TSSOP20 package.

## Chapter 3 Electrical Characteristics

### 3.1 Test Conditions

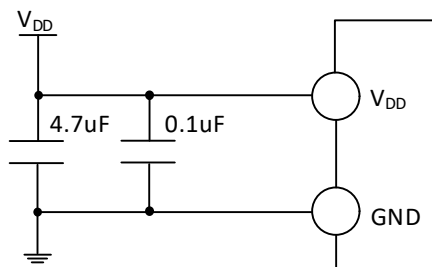
All voltages are referenced to GND unless otherwise stated and labelled.

All minimum and maximum values are guaranteed in the worst conditions of ambient temperature, supply voltage and clock frequency. Typical values are based on normal temperature (25°C) and  $V_{DD} = 5V$  environment, which are given only as design guidelines.

The data based on comprehensive evaluation, design simulation or technology characteristics are not tested in production. On the basis of comprehensive evaluation, the minimum and maximum values refer to sample tests. Unless otherwise specified that is tested, the characteristic parameters are guaranteed by comprehensive evaluation or design.

Power supply scheme:

Figure 3-1 Typical circuit for conventional power supply



### 3.2 Absolute Maximum Ratings

Critical or exceeding the absolute maximum value may cause the chip to operate improperly or even be damaged.

Table 3-1 Absolute maximum ratings

| Symbol               | Description                                                   | Min. | Max.         | Unit |
|----------------------|---------------------------------------------------------------|------|--------------|------|
| $T_A$                | Ambient temperature during operation                          | -40  | 85           | °C   |
| $T_S$                | Ambient temperature during storage                            | -40  | 125          | °C   |
| $V_{DD}$             | Voltage on external mains supply pin VDD                      | -0.3 | 6.0          | V    |
| $V_{IN}$             | Voltage on I/O pins                                           | -0.3 | $V_{DD}+0.3$ | V    |
| $ \Delta V_{DD\_x} $ | Voltage difference between each VDD of the main supply pins   |      | 20           | mV   |
| $ \Delta GND\_x $    | Voltage difference between each GND of the common ground pins |      | 20           | mV   |
| $V_{ESD(HBM)}$       | ESD electrostatic discharge voltage (HBM) on common I/O pins  | 4K   |              | V    |
| $I_{VDD}$            | Total combined current of all VDD main supply pins            |      | 150          | mA   |
| $I_{GND}$            | Total combined current on all GND common ground pins          |      | 200          | mA   |
| $I_{IO}$             | Sink current on any I/O pins                                  |      | 40           | mA   |
|                      | Source current on any I/O pins                                |      | 30           | mA   |



### 3.3 Electrical Characteristics

#### 3.3.1 Operating Conditions

Table 3-2 General operating conditions

| Symbol                  | Parameter                                                      | Condition                  | Min. | Max. | Unit |
|-------------------------|----------------------------------------------------------------|----------------------------|------|------|------|
| $F_{HCLK}$ or $F_{SYS}$ | Internal system bus frequency or microprocessor main frequency |                            |      | 48   | MHz  |
| $V_{DD}$                | Operating supply voltage (nominal 5V)                          | Disable USB or PD function | 2.0  | 5.5  | V    |
|                         |                                                                | Enable USB or PD function  | 3.0  | 5.3  |      |
|                         |                                                                | Disable ADC function       | 2.0  | 5.5  | V    |
|                         |                                                                | Enable ADC function        | 2.5  | 5.5  |      |

Table 3-3 Power-on and power-down conditions

| Symbol    | Parameter               | Condition | Min. | Max.     | Unit |
|-----------|-------------------------|-----------|------|----------|------|
| $t_{VDD}$ | $V_{DD}$ rise time rate |           | 0.1  | $\infty$ | us/V |
|           | $V_{DD}$ fall time rate |           | 10   | $\infty$ |      |

#### 3.3.2 Built-in Reset and Power Control Block Characteristics

Table 3-4 Reset and voltage monitor (For PDR, select high threshold gear)

| Symbol          | Parameter                                        | Condition                    | Min. | Typ. | Max. | Unit |
|-----------------|--------------------------------------------------|------------------------------|------|------|------|------|
| $V_{PVD}^{(1)}$ | Level selection of programmable voltage detector | PLS[1:0] = 00 (rising edge)  |      | 2.12 |      | V    |
|                 |                                                  | PLS[1:0] = 00 (falling edge) |      | 2.1  |      |      |
|                 |                                                  | PLS[1:0] = 01 (rising edge)  |      | 2.32 |      | V    |
|                 |                                                  | PLS[1:0] = 01 (falling edge) |      | 2.3  |      |      |
|                 |                                                  | PLS[1:0] = 10 (rising edge)  |      | 3.02 |      | V    |
|                 |                                                  | PLS[1:0] = 10 (falling edge) |      | 3    |      |      |
|                 |                                                  | PLS[1:0] = 11 (rising edge)  |      | 4.02 |      | V    |
|                 |                                                  | PLS[1:0] = 11 (falling edge) |      | 4    |      |      |
| $V_{PVDhyst}$   | PVD hysteresis                                   |                              |      | 20   |      | mV   |
| $V_{POR/PDR}$   | Power-on/power-down reset threshold              | Rising edge                  |      | 1.8  |      | V    |
|                 |                                                  | Falling edge                 |      | 1.78 |      | V    |
| $V_{PDRhyst}$   | PDR hysteresis                                   |                              |      | 20   |      | mV   |
| $t_{RSTTEMPO}$  | Power on reset                                   |                              | 4    | 17   | 24   | ms   |
|                 | Other resets                                     |                              | 6    | 9    | 20   | us   |

Note: 1. Normal temperature test value.

#### 3.3.3 Embedded Reference Voltage

Table 3-5 Embedded reference voltage

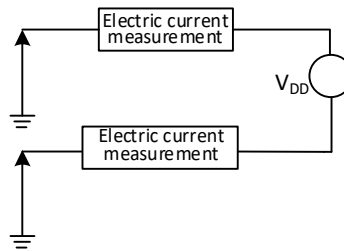
| Symbol           | Parameter                  | Condition                                           | Min. | Typ. | Max. | Unit        |
|------------------|----------------------------|-----------------------------------------------------|------|------|------|-------------|
| $V_{REFINT}$     | Internal reference voltage | $T_A = -40^{\circ}\text{C} \sim 85^{\circ}\text{C}$ | 1.16 | 1.2  | 1.24 | V           |
| $T_{S\_vrefint}$ | ADC sampling time when     | Slow sampling recommended                           |      |      | 11   | $1/f_{ADC}$ |

|  |                                        |  |  |  |  |  |
|--|----------------------------------------|--|--|--|--|--|
|  | reading the internal reference voltage |  |  |  |  |  |
|--|----------------------------------------|--|--|--|--|--|

### 3.3.4 Supply Current Characteristics

Current consumption is a comprehensive index of a variety of parameters and factors. These parameters and factors include operating voltage, ambient temperature, I/O pin load, software configuration of the product, the operating frequency, flip rate of the I/O pin, the location of the program in memory and the executed code, etc. The current consumption measurement method is as follows:

Figure 3-2 Current consumption measurement



The microcontroller is in the following conditions:

When tested at room temperature  $V_{DD} = 3.3V$ : all I/O ports configured with pull-up inputs, HSI = 48M. power consumption of all peripheral clocks enabled or disabled.

Table 3-6 Typical current consumption in Run mode, the data processing code runs from the internal Flash

| Symbol                         | Parameter           | Condition                                                                                        |                           | Typ.                   |                         | Unit |
|--------------------------------|---------------------|--------------------------------------------------------------------------------------------------|---------------------------|------------------------|-------------------------|------|
|                                |                     |                                                                                                  |                           | Enable all peripherals | Disable all peripherals |      |
| I <sub>DD</sub> <sup>(1)</sup> | Current in Run mode | Runs on the high-speed internal RC oscillator (HSI). Uses AHB prescaler to reduce the frequency. | F <sub>HCLK</sub> = 48MHz | 4.2                    | 3.0                     | mA   |
|                                |                     |                                                                                                  | F <sub>HCLK</sub> = 24MHz | 3.2                    | 2.6                     |      |
|                                |                     |                                                                                                  | F <sub>HCLK</sub> = 16MHz | 2.5                    | 2.1                     |      |
|                                |                     |                                                                                                  | F <sub>HCLK</sub> = 8MHz  | 2.2                    | 2.0                     |      |

Note: The above are measured parameters.

Table 3-7 Typical current consumption in Sleep mode, data processing code runs from internal Flash or SRAM

| Symbol                         | Parameter                                                                                 | Condition                                                                                           |                           | Typ.                   |                         | Unit |
|--------------------------------|-------------------------------------------------------------------------------------------|-----------------------------------------------------------------------------------------------------|---------------------------|------------------------|-------------------------|------|
|                                |                                                                                           |                                                                                                     |                           | Enable all peripherals | Disable all peripherals |      |
| I <sub>DD</sub> <sup>(1)</sup> | Current in Sleep mode<br>(In this case, peripheral power supply and clock are maintained) | Runs on the high-speed internal RC oscillator (HSI).<br>Uses AHB prescaler to reduce the frequency. | F <sub>HCLK</sub> = 48MHz | 3.0                    | 1.8                     | mA   |
|                                |                                                                                           |                                                                                                     | F <sub>HCLK</sub> = 24MHz | 2.1                    | 1.5                     |      |
|                                |                                                                                           |                                                                                                     | F <sub>HCLK</sub> = 16MHz | 1.8                    | 1.4                     |      |
|                                |                                                                                           |                                                                                                     | F <sub>HCLK</sub> = 8MHz  | 1.5                    | 1.3                     |      |

Note: The above are measured parameters.

Table 3-8 Typical current consumption in Stop and Standby mode

| Symbol          | Parameter               | Condition                                                          | Typ. | Unit |
|-----------------|-------------------------|--------------------------------------------------------------------|------|------|
| I <sub>DD</sub> | Current in Stop mode    | High-speed internal RC oscillator is off (no independent watchdog) | 75   | uA   |
|                 | Current in Standby mode | Independent watchdog enabled                                       | 530  |      |
|                 |                         | Independent watchdog disabled                                      | 54   |      |

Note: The above are measured parameters.

### 3.3.5 Internal Clock Source Characteristics

Table 3-9 Internal high-speed (HSI) RC oscillator characteristics

| Symbol               | Parameter                                      | Condition       | Min. | Typ. | Max. | Unit |
|----------------------|------------------------------------------------|-----------------|------|------|------|------|
| F <sub>HSI</sub>     | Frequency (after calibration)                  |                 |      | 48   |      | MHz  |
| DuCy <sub>HSI</sub>  | Duty cycle                                     |                 | 45   | 50   | 55   | %    |
| ACC <sub>HSI</sub>   | Accuracy of HSI oscillator (after calibration) | TA = 0°C~70°C   | -1.7 | ±0.8 | 1.6  | %    |
|                      |                                                | TA = -40°C~85°C | -2.6 | ±1.1 | 2.2  | %    |
| t <sub>SU(HSI)</sub> | HSI oscillator startup stabilization time      |                 | 1.5  |      | 3.5  | us   |
| I <sub>DD(HSI)</sub> | HSI oscillator power consumption               |                 |      | 312  |      | uA   |

### 3.3.6 Wakeup Time from Low-power Mode

Table 3-10 Wakeup time from low-power mode<sup>(1)</sup>

| Symbol               | Parameter                | Condition                  | Typ. | Unit |
|----------------------|--------------------------|----------------------------|------|------|
| t <sub>wusleep</sub> | Wakeup from Sleep mode   | Wake up using HSI RC clock | 1    | us   |
| t <sub>wustop</sub>  | Wakeup from Stop mode    | Wake up using HSI RC clock | 10   | us   |
| t <sub>wustdb</sub>  | Wakeup from Standby mode | Wake up using HSI RC clock | 10   | us   |

Note: The above parameters are measured parameters.

### 3.3.7 Memory Characteristics

Table 3-11 Flash memory characteristics

| Symbol                  | Parameter                         | Condition | Min. | Typ. | Max. | Unit |
|-------------------------|-----------------------------------|-----------|------|------|------|------|
| t <sub>prog_page</sub>  | Page (256 bytes) programming time |           |      | 1.5  | 2.0  | ms   |
| t <sub>erase_page</sub> | Page (256 bytes) erase time       |           |      | 2.5  | 3.0  | ms   |
| t <sub>erase_sec</sub>  | Sector (1K bytes) erase time      |           |      | 2.7  | 3.3  | ms   |

Table 3-12 Flash memory endurance and data retention

| Symbol           | Parameter             | Condition             | Min. | Typ. | Max. | Unit  |
|------------------|-----------------------|-----------------------|------|------|------|-------|
| N <sub>END</sub> | Endurance             | T <sub>A</sub> = 25°C | 100K |      |      | Times |
| t <sub>RET</sub> | Data retention period |                       | 10   |      |      | Years |

### 3.3.8 I/O Port Characteristics

Table 3-13 General-purpose I/O static characteristics

| Symbol    | Parameter                                             | Condition                         | Min.                       | Typ. | Max.                       | Unit |
|-----------|-------------------------------------------------------|-----------------------------------|----------------------------|------|----------------------------|------|
| $V_{IH}$  | Standard I/O pin, input high-level voltage            |                                   | $(V_{DD}-2)^*$<br>0.36+1.3 |      | $V_{DD}$                   | V    |
|           |                                                       | $V_{DD} = 5V$                     | 2.4                        |      | $V_{DD}$                   | V    |
|           |                                                       | $V_{DD} = 3.3V$                   | 1.8                        |      | $V_{DD}$                   | V    |
| $V_{IL}$  | Standard I/O pin, input low-level voltage             |                                   | 0                          |      | $(V_{DD}-2)^*$<br>0.24+0.4 | V    |
|           |                                                       | $V_{DD} = 5V$                     | 0                          |      | 1.1                        | V    |
|           |                                                       | $V_{DD} = 3.3V$                   | 0                          |      | 0.7                        | V    |
| $V_{OH}$  | Standard I/O pin, output high-level voltage           | $I_{IO} = 6mA$<br>$V_{DD} = 3.3V$ | $V_{DD}-0.4$               |      |                            | V    |
|           |                                                       | $I_{IO} = 12mA$<br>$V_{DD} = 5V$  | $V_{DD}-0.5$               |      |                            | V    |
| $V_{OL}$  | Standard I/O pin, output low-level voltage            | $I_{IO} = 8mA$<br>$V_{DD} = 3.3V$ |                            |      | 0.4                        | V    |
|           |                                                       | $I_{IO} = 16mA$<br>$V_{DD} = 5V$  |                            |      | 0.5                        | V    |
| $V_{hys}$ | Standard I/O pin Schmitt trigger voltage hysteresis   | $V_{DD} = 5V$                     | 180                        | 350  |                            | mV   |
| $I_{lkg}$ | Input leakage current                                 |                                   | -2                         |      | 2                          | uA   |
| $I_{PU}$  | Standard I/O pin weak pull-up current                 | $V_{DD} = 5V$                     | 25                         | 60   | 140                        | uA   |
|           |                                                       | $V_{DD} = 3.3V$                   | 12                         | 30   | 65                         | uA   |
| $I_{PD}$  | PA0-PA15 pins weak pull-down current                  | $V_{DD} = 5V$                     | 60                         | 150  | 350                        | uA   |
|           |                                                       | $V_{DD} = 3.3V$                   | 30                         | 75   | 180                        | uA   |
| $C_{IO}$  | Single I/O pin capacitor (without double I/O co-seal) |                                   |                            | 5    |                            | pF   |

Note: The above are guaranteed design parameters;

#### Output Drive Current Characteristics

The GPIOs (General-purpose Input/Output) can absorb or output up to  $\pm 8mA$  of current. In user applications, the total current driven by all IO pins must not exceed the absolute maximum ratings given in section 3.2.

Table 3-14 Input/output AC characteristics

| Pin | Symbol           | Parameter                    | Condition                      | Min. | Max. | Unit |
|-----|------------------|------------------------------|--------------------------------|------|------|------|
| PA  | $F_{max(IO)out}$ | Maximum frequency            | $CL=50pF, V_{DD}=2.9\sim 4.0V$ |      | 40   | MHz  |
|     |                  |                              | $CL=50pF, V_{DD}=4.0\sim 5.5V$ |      | 56   | MHz  |
|     | $t_{f(IO)out}$   | Output high to low fall time | $CL=50pF, V_{DD}=2.9\sim 4.0V$ |      | 6    | ns   |
|     |                  |                              | $CL=50pF, V_{DD}=4.0\sim 5.5V$ |      | 4.2  | ns   |
|     | $t_{r(IO)out}$   | Output low to high rise time | $CL=50pF, V_{DD}=2.9\sim 4.0V$ |      | 8.4  | ns   |
|     |                  |                              | $CL=50pF, V_{DD}=4.0\sim 5.5V$ |      | 6    | ns   |
| PB  | $F_{max(IO)out}$ | Maximum frequency            | $CL=50pF, V_{DD}=2.9\sim 4.0V$ |      | 16   | MHz  |

|    |                  |                              |                             |      |     |
|----|------------------|------------------------------|-----------------------------|------|-----|
| PC | $t_{f(IO)out}$   | Output high to low fall time | CL=50pF, $V_{DD}$ =4.0~5.5V | 24   | MHz |
|    |                  |                              | CL=50pF, $V_{DD}$ =2.9~4.0V | 6    | ns  |
|    | $t_{r(IO)out}$   | Output low to high rise time | CL=50pF, $V_{DD}$ =4.0~5.5V | 4.2  | ns  |
|    |                  |                              | CL=50pF, $V_{DD}$ =2.9~4.0V | 18   | ns  |
|    | $F_{max(IO)out}$ | Maximum frequency            | CL=50pF, $V_{DD}$ =2.9~4.0V | 13.2 | ns  |
|    |                  |                              | CL=50pF, $V_{DD}$ =4.0~5.5V | 28   | MHz |
|    | $t_{f(IO)out}$   | Output high to low fall time | CL=50pF, $V_{DD}$ =2.9~4.0V | 36   | MHz |
|    |                  |                              | CL=50pF, $V_{DD}$ =4.0~5.5V | 8.4  | ns  |
| PC | $t_{r(IO)out}$   | Output low to high rise time | CL=50pF, $V_{DD}$ =4.0~5.5V | 7.2  | ns  |
|    |                  |                              | CL=50pF, $V_{DD}$ =2.9~4.0V | 13.2 | ns  |
|    | $t_{f(IO)out}$   | Output high to low fall time | CL=50pF, $V_{DD}$ =2.9~4.0V | 9.6  | ns  |
|    |                  |                              | CL=50pF, $V_{DD}$ =4.0~5.5V |      |     |

Note: The above are guaranteed design parameters.

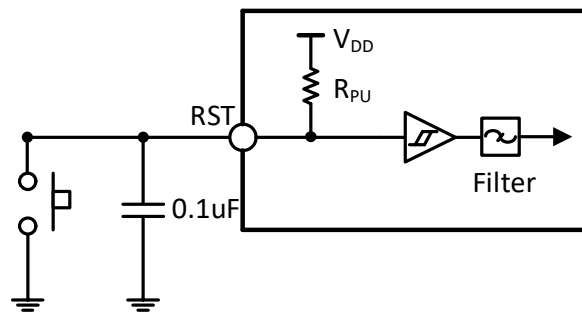
### 3.3.9 RST Pin Characteristics

Table 3-15 External reset pin characteristics

| Symbol       | Parameter             | Condition | Min. | Typ. | Max. | Unit |
|--------------|-----------------------|-----------|------|------|------|------|
| $V_{F(RST)}$ | RST input pulse width |           | 300  |      |      | ns   |

Circuit reference design and requirements:

Figure 3-3 Typical circuit for external reset pin



### 3.3.10 USB PD Interface Characteristics

Table 3-16 PD interface characteristics, application: PD communication

| Symbol      | Parameter                           | Condition                                                                    | Min. | Typ. | Max. | Unit |
|-------------|-------------------------------------|------------------------------------------------------------------------------|------|------|------|------|
| $t_{Rise}$  | Rising time                         | Time between 10% and 90% of the range, minimum value for no-load conditions. | 300  |      | 600  | ns   |
| $t_{Fall}$  | Falling time                        | Time between 10% and 90% of the range, minimum value for no-load conditions. | 300  |      | 600  | ns   |
| $V_{Swing}$ | Output voltage swing (peak-to-peak) | Low voltage output mode, CL=50pF                                             | 1.04 | 1.12 | 1.20 | V    |
| $I_{pu}$    | CC pull-up current                  | Pin voltage < $V_{DD} - 1V$ , PUCC[1:0] = 11                                 | 64   | 80   | 96   | uA   |
|             |                                     | Pin voltage < $V_{DD} - 1V$ , PUCC[1:0] = 10                                 | 144  | 180  | 216  | uA   |
|             |                                     | Pin voltage < $V_{DD} - 1V$ , PUCC[1:0] = 01                                 | 264  | 330  | 396  | uA   |

### 3.3.11 TIM Timer Characteristics

Table 3-17 TIMx characteristics

| Symbol           | Parameter                                                      | Condition             | Min.   | Max.            | Unit                |
|------------------|----------------------------------------------------------------|-----------------------|--------|-----------------|---------------------|
| $t_{res(TIM)}$   | Timer reference clock                                          |                       | 1      |                 | $t_{TIMxCLK}$<br>LK |
|                  |                                                                | $f_{TIMxCLK} = 48MHz$ | 20.8   |                 | ns                  |
| $F_{EXT}$        | Timer external clock frequency on CH1 to CH4                   |                       | 0      | $f_{TIMxCLK}/2$ | MHz                 |
|                  |                                                                | $f_{TIMxCLK} = 48MHz$ | 0      | 24              | MHz                 |
| $R_{esTIM}$      | Timer resolution                                               |                       |        | 16              | Bit                 |
| $t_{COUNTER}$    | 16-bit counter clock cycle when the internal clock is selected |                       | 1      | 65536           | $t_{TIMxCLK}$<br>LK |
|                  |                                                                | $f_{TIMxCLK} = 48MHz$ | 0.0208 | 1363            | us                  |
| $t_{MAX\_COUNT}$ | Maximum possible count                                         |                       |        | 65535           | $t_{TIMxCLK}$<br>LK |
|                  |                                                                | $f_{TIMxCLK} = 48MHz$ |        | 1363            | us                  |

### 3.3.12 I2C Interface Characteristics

Figure 3-4 I2C bus timing diagram

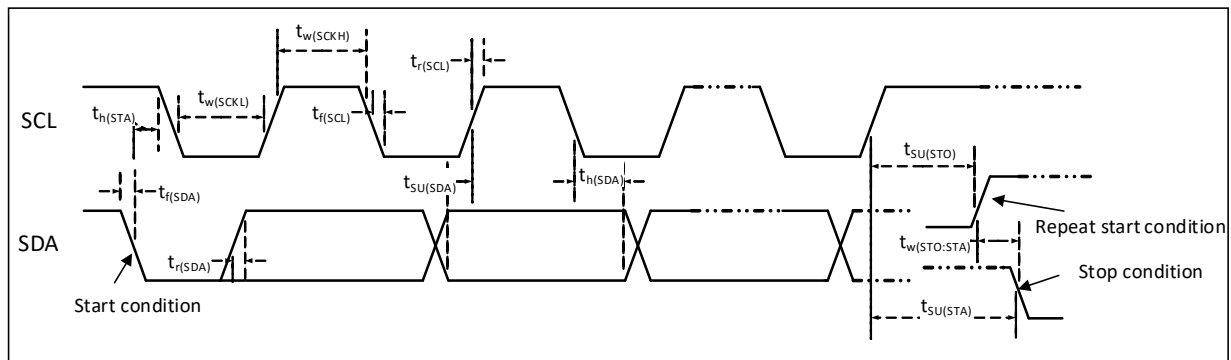


Table 3-18 I2C interface characteristics

| Symbol              | Parameter                                   | Standard I2C |      | Fast I2C |      | Unit |
|---------------------|---------------------------------------------|--------------|------|----------|------|------|
|                     |                                             | Min.         | Max. | Min.     | Max. |      |
| $t_w(SCKL)$         | SCL clock low level time                    | 4.7          |      | 1.2      |      | us   |
| $t_w(SCKH)$         | SCL clock high level time                   | 4.0          |      | 0.6      |      | us   |
| $t_{SU}(SDA)$       | SDA data setup time                         | 250          |      | 100      |      | ns   |
| $t_H(SDA)$          | SDA data hold time                          | 0            |      | 0        | 900  | ns   |
| $t_r(SDA)/t_r(SCL)$ | SDA and SCL rise time                       |              | 1000 | 20       |      | ns   |
| $t_f(SDA)/t_f(SCL)$ | SDA and SCL fall time                       |              | 300  |          |      | ns   |
| $t_H(STA)$          | Start condition hold time                   | 4.0          |      | 0.6      |      | us   |
| $t_{SU}(STA)$       | Repeated start condition setup time         | 4.7          |      | 0.6      |      | us   |
| $t_{SU}(STO)$       | Stop condition setup time                   | 4.0          |      | 0.6      |      | us   |
| $t_w(STO:STA)$      | Time from stop condition to start condition | 4.7          |      | 1.2      |      | us   |

|       |                              |  |     |  |     |    |
|-------|------------------------------|--|-----|--|-----|----|
|       | (bus free)                   |  |     |  |     |    |
| $C_b$ | Capacitive load for each bus |  | 400 |  | 400 | pF |

### 3.3.13 SPI Interface Characteristics

Figure 3-5 SPI timing diagram in Master mode

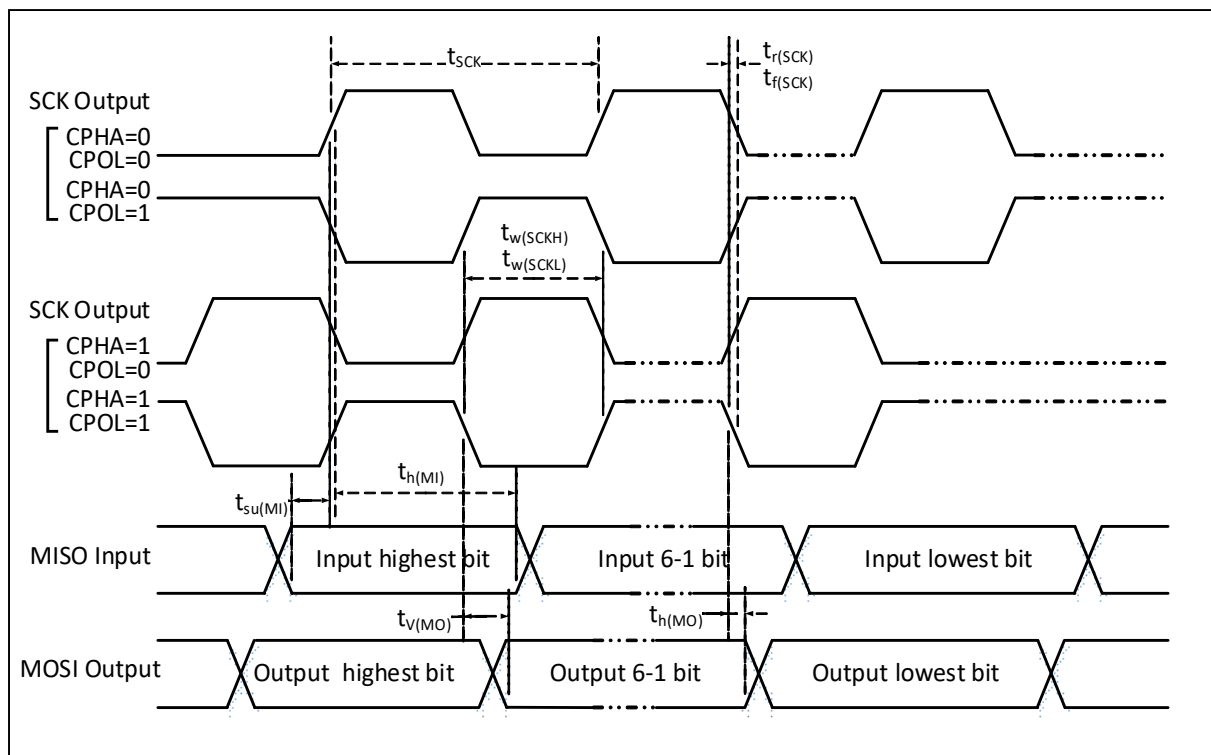


Figure 3-6 SPI timing diagram in Slave mode (CPHA=0)

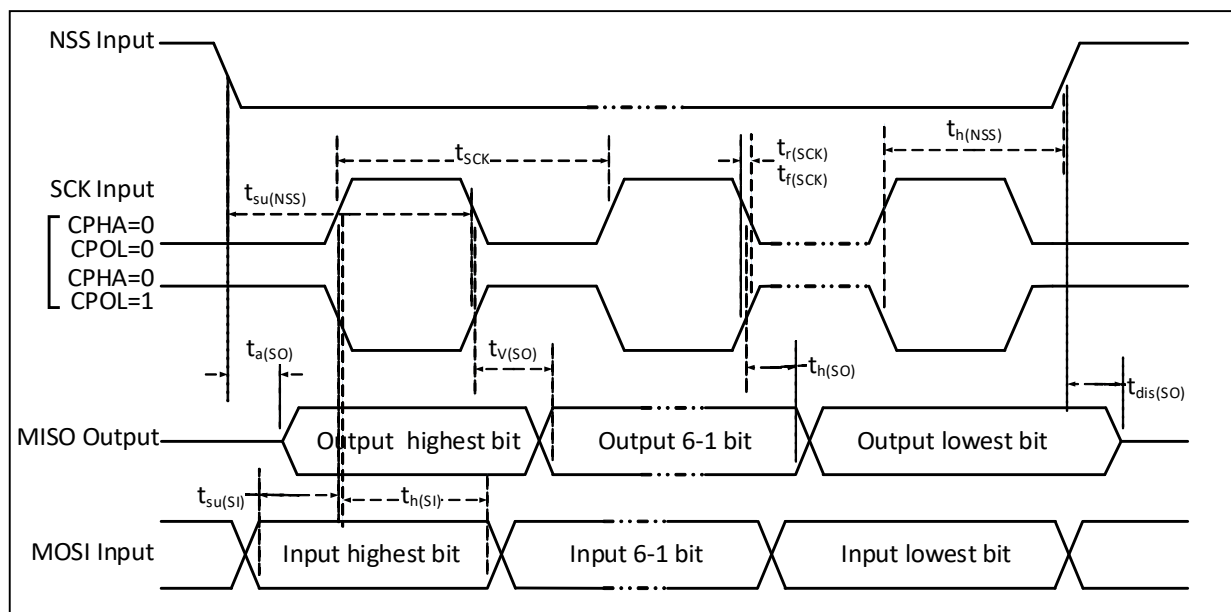


Figure 3-7 SPI timing diagram in Slave mode (CPHA=1)

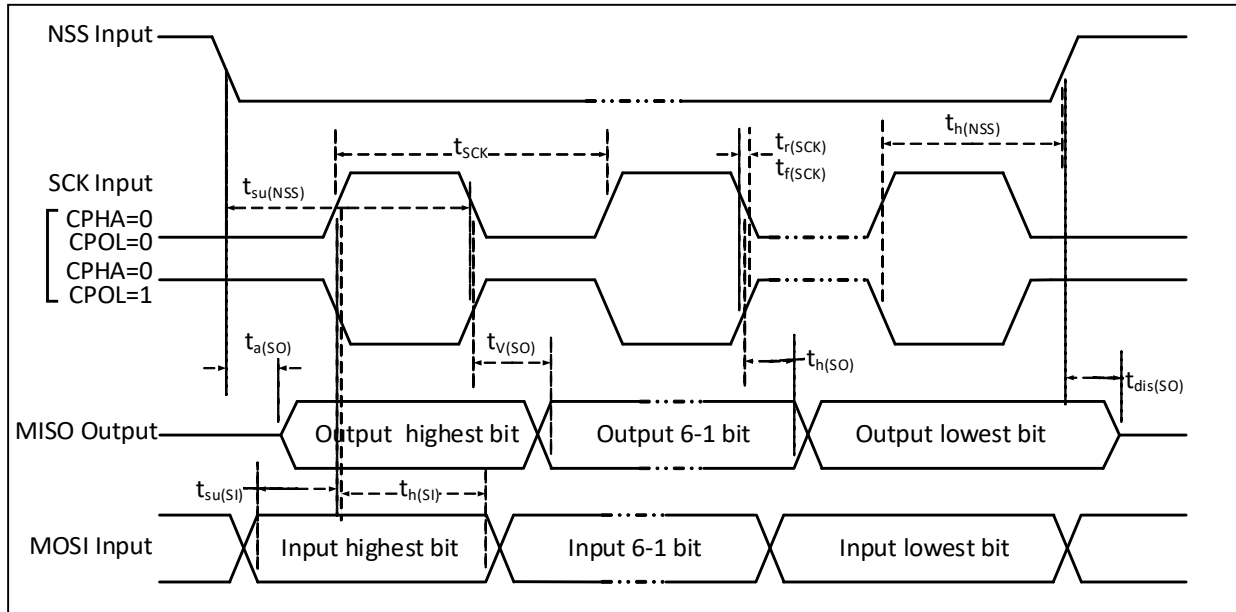


Table 3-19 SPI interface characteristics

| Symbol                    | Parameter                         | Condition                                              | Min.        | Max.        | Unit |
|---------------------------|-----------------------------------|--------------------------------------------------------|-------------|-------------|------|
| $f_{SCK}/t_{SCK}$         | SPI clock frequency               | Master mode                                            |             | 24          | MHz  |
|                           |                                   | Slave mode                                             |             | 24          | MHz  |
| $t_{r(SCK)}/t_{f(SCK)}$   | SPI clock rise and fall time      | Load capacitance: C = 30pF                             |             | 20          | ns   |
| $t_{su(NSS)}$             | NSS setup time                    | Slave mode                                             | $2t_{HCLK}$ |             | ns   |
| $t_{h(NSS)}$              | NSS hold time                     | Slave mode                                             | $2t_{HCLK}$ |             | ns   |
| $t_{w(SCKH)}/t_{w(SCKL)}$ | SCK high-level and low-level time | Master mode, $f_{PCLK} = 24MHz$ , Prescaler factor = 4 | 70          | 100         | ns   |
| $t_{su(MI)}$              | Data input setup time             | Master mode                                            | 5           |             | ns   |
| $t_{su(SI)}$              |                                   | Slave mode                                             | 5           |             | ns   |
| $t_{h(MI)}$               | Data input hold time              | Master mode                                            | 5           |             | ns   |
| $t_{h(SI)}$               |                                   | Slave mode                                             | 4           |             | ns   |
| $t_{a(SO)}$               | Data output access time           | Slave mode, $f_{PCLK} = 20MHz$                         | 0           | $1t_{HCLK}$ | ns   |
| $t_{dis(SO)}$             | Data output disable time          | Slave mode                                             | 0           | 10          | ns   |
| $t_{v(SO)}$               | Data output valid time            | Slave mode (After enable edge)                         |             | 25          | ns   |
| $t_{v(MO)}$               |                                   | Master mode (After enable edge)                        |             | 5           | ns   |
| $t_{h(SO)}$               | Data output hold time             | Slave mode (After enable edge)                         | 15          |             | ns   |
| $t_{h(MO)}$               |                                   | Master mode (After enable edge)                        | 0           |             | ns   |

### 3.3.14 USB Interface Characteristics

Table 3-20 USB interface I/O characteristics

| Symbol   | Parameter                       | Condition                                                 | Min. | Typ. | Max. | Unit |
|----------|---------------------------------|-----------------------------------------------------------|------|------|------|------|
| $V_{DD}$ | USB operating voltage           | Selection of USB parameters according to $V_{DD}$ voltage | 3.0  |      | 5.3  | V    |
| $V_{SE}$ | Single-ended receiver threshold | Nominal voltage                                           | 1.2  |      | 1.9  | V    |



|               |                                 |  |     |     |     |   |
|---------------|---------------------------------|--|-----|-----|-----|---|
| $V_{OL}$      | Static output low level         |  |     |     | 0.3 | V |
| $V_{OH}$      | Static output high level        |  | 2.8 |     |     | V |
| $V_{BC\_REF}$ | BC comparator reference voltage |  |     | 0.4 |     | V |
| $V_{BC\_SRC}$ | BC protocol output voltage      |  |     | 0.6 |     | V |

### 3.3.15 12-bit ADC Characteristics

Table 3-21 ADC characteristics

| Symbol     | Parameter                                       | Condition                  | Min. | Typ. | Max.     | Unit         |
|------------|-------------------------------------------------|----------------------------|------|------|----------|--------------|
| $V_{DD}$   | Supply voltage                                  | Rated performance          | 3.0  | 5    | 5.3      | V            |
|            |                                                 | Performance may be reduced | 2.5  |      | 5.5      | V            |
| $I_{DD}$   | Supply current                                  |                            |      | 290  | 480      | uA           |
| $f_{ADC}$  | ADC clock frequency                             | $V_{DD} \geq 3.2V$         | 3    |      | 8        | MHz          |
|            |                                                 | $V_{DD} < 3.2V$            | 3    |      | 6        | MHz          |
| $f_s$      | Sampling rate                                   | $V_{DD} \geq 3.2V$         | 125  |      | 470      | KHz          |
|            |                                                 | $V_{DD} < 3.2V$            | 125  |      | 353      | KHz          |
| $V_{AIN}$  | Conversion voltage range                        |                            | 0    |      | $V_{DD}$ | V            |
| $R_{ADC}$  | Sampling switch resistance                      |                            | 0.5  | 0.6  | 1.5      | k $\Omega$   |
| $C_{ADC}$  | Internal sample and hold capacitor              |                            |      | 21   |          | pF           |
| $t_{lat}$  | Injected trigger conversion latency             |                            |      | 1    |          | 1/ $f_{ADC}$ |
| $t_{latr}$ | Regular trigger conversion latency              |                            |      | 1    |          | 1/ $f_{ADC}$ |
| $t_s$      | Sampling time                                   |                            |      | 3.5  |          | 1/ $f_{ADC}$ |
| $t_{CONV}$ | Total conversion time (including sampling time) |                            | 17   |      | 24       | 1/ $f_{ADC}$ |

Note: The above are guaranteed design parameters.

Table 3-22 ADC error

| Symbol | Parameter                       | Condition                                                                      | Min. | Typ.    | Max.     | Unit |
|--------|---------------------------------|--------------------------------------------------------------------------------|------|---------|----------|------|
| EO     | Offset error                    | $f_{ADC} = 3\text{ MHz}$ ,<br>$R_{AIN} < 10\text{ k}\Omega$ ,<br>$V_{DD} = 5V$ |      | $\pm 4$ |          | LSB  |
| ED     | Differential nonlinearity error |                                                                                |      | $\pm 1$ | $\pm 10$ |      |
| EL     | Integral nonlinearity error     |                                                                                |      | $\pm 4$ | $\pm 20$ |      |

Note: The above are guaranteed design parameters.

$C_p$  represents the parasitic capacitance on the PCB and the pad (about 5pF), which may be related to the quality of the pad and PCB layout. A larger  $C_p$  value will reduce the conversion accuracy, the solution is to reduce the  $f_{ADC}$  value.

Figure 3-8 ADC typical connection diagram

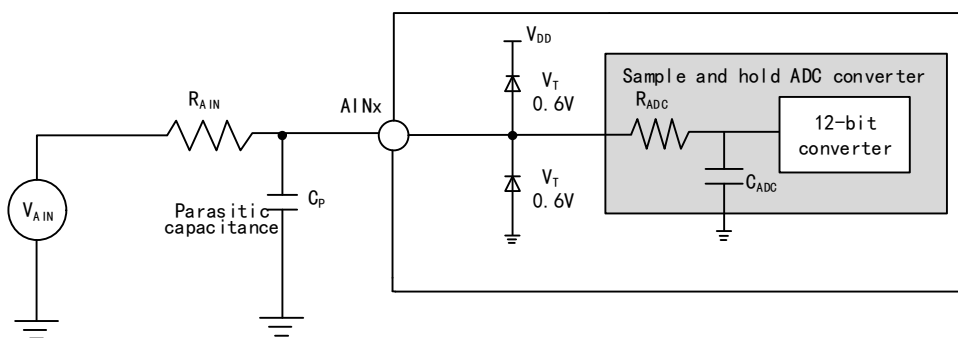
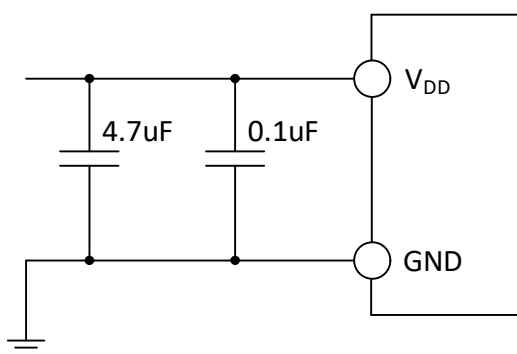


Figure 3-9 Analog power supply and decoupling circuit reference



### 3.3.16 OPA Characteristics

Table 3-23 OPA characteristics

| Symbol          | Parameter                    | Condition: $V_{DD} = 5V$                   | Min. | Typ. | Max.     | Unit |
|-----------------|------------------------------|--------------------------------------------|------|------|----------|------|
| $V_{DD}$        | Supply voltage               | Recommended not less than 2.5V             | 2    | 5    | 5.5      | V    |
| $V_{CM}$        | Common mode input voltage    |                                            | 0    |      | $V_{DD}$ | V    |
| $V_{IOFFSET}$   | Input offset voltage         | Common mode input $V_{CM} = 0.5V$          |      | 5    | 13       | mV   |
|                 |                              | Common mode input $V_{CM} = V_{DD}/2$      |      | 3    | 10       |      |
|                 |                              | Common mode input $V_{CM} = V_{DD} - 0.5V$ |      | 5    | 13       |      |
| $I_{LOAD}$      | Drive current                | $R_{LOAD} = 5k\Omega$                      |      |      | 1        | mA   |
| $I_{LOAD\_PGA}$ | PGA mode drive current       |                                            |      |      | 400      | uA   |
| $I_{DDOPAMP}$   | Current consumption          | No load, static mode                       |      | 210  |          | uA   |
| $CMRR^{(1)}$    | Common mode rejection ratio  | @1kHz                                      |      | 110  |          | dB   |
| $PSRR^{(1)}$    | Power supply rejection ratio | @1kHz                                      |      | 71   |          | dB   |
| $A_v^{(1)}$     | Open loop gain               | $C_{LOAD} = 5pF$                           |      | 110  |          | dB   |
| $G_{BW}^{(1)}$  | Unit gain bandwidth          | $C_{LOAD} = 5pF$                           |      | 13   |          | MHz  |
| $P_M^{(1)}$     | Phase margin                 | $C_{LOAD} = 5pF$                           |      | 88   |          |      |
| $S_R^{(1)}$     | Slew rate limited            | $C_{LOAD} = 5pF$                           |      | 5    |          | V/us |

|                            |                                           |                                                                 |              |     |    |                      |
|----------------------------|-------------------------------------------|-----------------------------------------------------------------|--------------|-----|----|----------------------|
| $t_{WAKUP}^{(1)}$          | Setup time from shutdown to wake up, 0.1% | Input $V_{DD}/2$ ,<br>$C_{LOAD} = 50pF$ , $R_{LOAD} = 5k\Omega$ |              |     | 1  | us                   |
| $R_{LOAD}$                 | Resistive load                            |                                                                 | 5            |     |    | k $\Omega$           |
| $C_{LOAD}$                 | Capacitive load                           |                                                                 |              |     | 50 | pF                   |
| $V_{OHSAT}^{(2)}$          | High saturation output voltage            | $R_{LOAD} = 5k\Omega$                                           | $V_{DD}-300$ |     |    | mV                   |
|                            |                                           | $R_{LOAD} = 20k\Omega$                                          | $V_{DD}-50$  |     |    |                      |
| $V_{OLSAT}^{(2)}$          | Low saturation output voltage             | $R_{LOAD} = 5k\Omega$                                           |              |     | 10 | mV                   |
|                            |                                           | $R_{LOAD} = 20k\Omega$                                          |              |     | 7  |                      |
| PGA<br>Gain <sup>(1)</sup> | NSEL=010b mode in phase                   | Gain =16, PA1=GND                                               | -3           |     | 3  | %                    |
|                            | Internal in-phase PGA                     | Gain = 4<br>$V_{INP} < (V_{DD}/7)$                              | -1           |     | 1  | %                    |
|                            |                                           | Gain = 8<br>$V_{INP} < (V_{DD}/15)$                             | -1           |     | 1  | %                    |
|                            |                                           | Gain = 16<br>$V_{INP} < (V_{DD}/31)$                            | -1           |     | 1  | %                    |
|                            |                                           | Gain = 32<br>$V_{INP} < (V_{DD}/63)$                            | -1           |     | 1  | %                    |
|                            |                                           |                                                                 |              |     |    |                      |
| Delta R                    | Absolute change in resistance             |                                                                 | -15          |     | 15 | %                    |
| $EN^{(1)}$                 | Equivalent input voltage noise            | $R_{LOAD} = 5k\Omega @ 1kHz$                                    |              | 100 |    | nV/<br>sqrt(Hz)<br>) |
|                            |                                           | $R_{LOAD} = 20k\Omega @ 1KHz$                                   |              | 60  |    |                      |

Note: 1. The source simulation is not a real measurement.

2. The load current limits the saturated output voltage.

### 3.3.17 CMP Characteristics

Table 3-24 CMP voltage comparator characteristics

| Symbol              | Parameter                                                                                    | Condition: $V_{DD} = 5V$       | Min. | Typ. | Max.     | Unit |
|---------------------|----------------------------------------------------------------------------------------------|--------------------------------|------|------|----------|------|
| $V_{DD}$            | Supply voltage                                                                               | Recommended not less than 2.5V | 2    | 5    | 5.5      | V    |
| $V_{CM}$            | Common mode input voltage                                                                    |                                | 0    |      | $V_{DD}$ | V    |
| $V_{IOFFSET}^{(1)}$ | Input offset voltage                                                                         |                                |      | 5    | 18       | mV   |
| $I_{DDOPAMP}$       | Current consumption                                                                          |                                |      | 75   |          | uA   |
| $t_D^{(1)}$         | Comparator delay.<br>$V_{INP}$ varies from ( $V_{INN}-100mV$ ) to ( $V_{INN}+100mV$ ) change | $0 \leq V_{INN} \leq V_{DD}$   |      | 15   | 50       | ns   |

Note: 1. Design parameters are guaranteed.

## Chapter 4 Package and Ordering Information

### Packages

| Part No.     | Package form | Body size | Pin pitch | Package description               | Packing type |
|--------------|--------------|-----------|-----------|-----------------------------------|--------------|
| CH32X035R8T6 | LQFP64M      | 10*10mm   | 0.5mm     | Low Profile Quad Flat Pack        | Tray         |
| CH32X035C8T6 | LQFP48       | 7*7mm     | 0.5mm     | Low Profile Quad Flat Pack        | Tray         |
| CH32X035G8U6 | QFN28        | 4*4mm     | 0.4mm     | Quad Flat No-Lead Package         | Tray         |
| CH32X035G8R6 | QSOP28       | 3.9*9.9mm | 0.635mm   | Quarter-sized Outline Package     | Tube         |
| CH32X035F8U6 | QFN20        | 3*3mm     | 0.4mm     | Quad Flat No-Lead Package         | Tape & Reel  |
| CH32X035F7P6 | TSSOP20      | 4.4*6.5mm | 0.65mm    | Thin Shrink Small Outline Package | Tube         |
| CH32X033F8P6 | TSSOP20      | 4.4*6.5mm | 0.65mm    | Thin Shrink Small Outline Package | Tube         |

Note: 1. The packing type of QFP/QFN is usually tray.

2. Size of tray: The size of Tray is generally a uniform size (322.6\*135.9\*7.62). There are differences in the size of the restriction holes for different package types, and there are differences between different packaging factories for tubes, please confirm with the manufacturer for details.

Note: All dimensions are in millimeters. The pin center spacing values are nominal values, with no error. Other than that, the dimensional error is not greater than the greater of  $\pm 0.2\text{mm}$  or 10%.

Figure 4-1 LQFP64M package

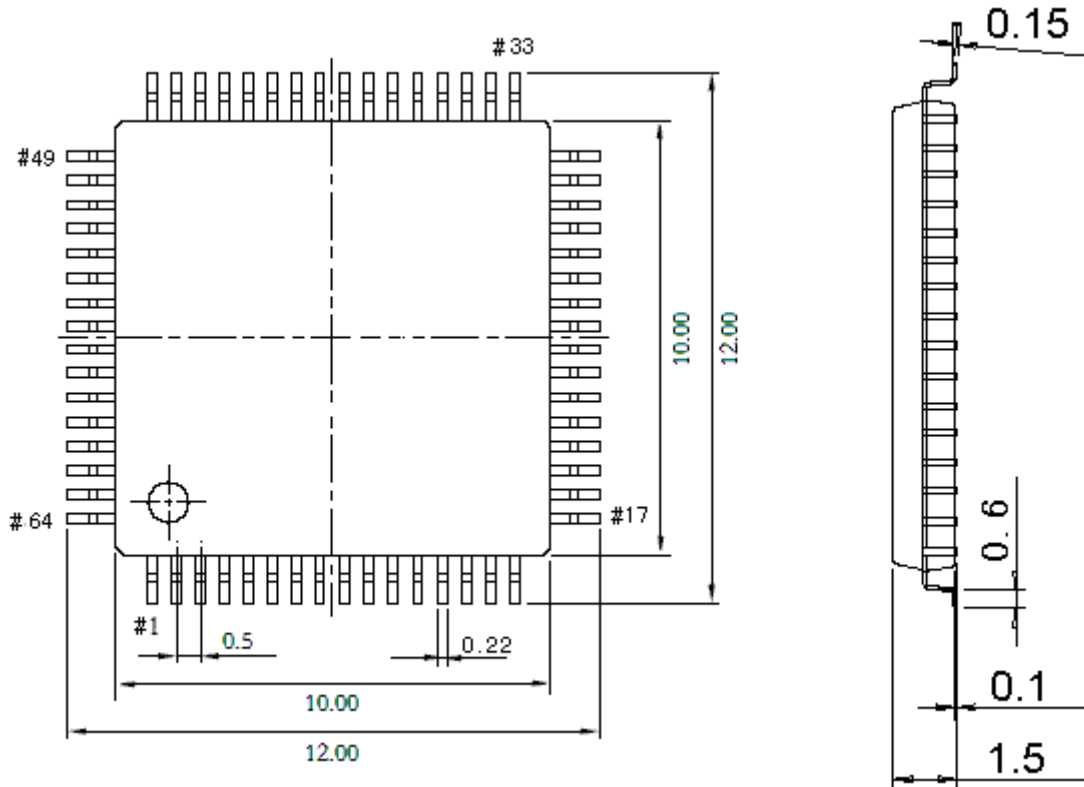


Figure 4-2 LQFP48 package

Figure 4-3 QFN28 package

Figure 4-4 QSOP28 package

Figure 4-5 QFN20 package

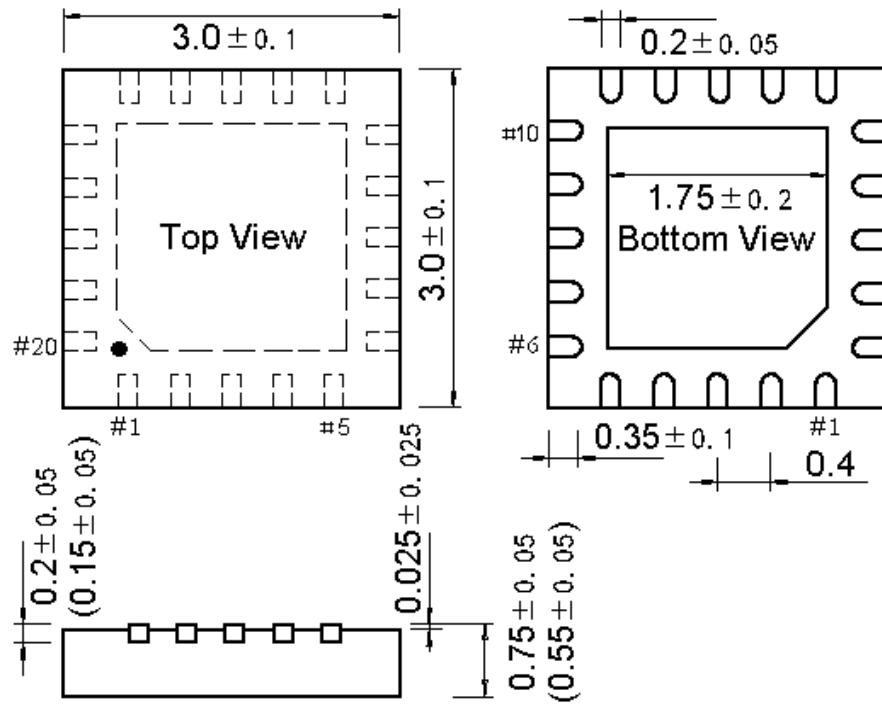


Figure 4-6 TSSOP20 package



## Series Product Naming Rules

|                                                                                     |              |              |             |         |   |   |   |   |
|-------------------------------------------------------------------------------------|--------------|--------------|-------------|---------|---|---|---|---|
| Example:                                                                            | CH32         | V            | 3           | 03      | R | 8 | T | 6 |
| Device family                                                                       |              |              |             |         |   |   |   |   |
| F = ARM-based, general-purpose MCU                                                  |              |              |             |         |   |   |   |   |
| V = QingKe RISC-V-based, general-purpose MCU                                        |              |              |             |         |   |   |   |   |
| L = QingKe RISC-V-based, low power MCU                                              |              |              |             |         |   |   |   |   |
| X = QingKe RISC-V-based, Dedicated architecture or special IO                       |              |              |             |         |   |   |   |   |
| Product type                                                                        |              |              |             |         |   |   |   |   |
| 0 = QingKe V2/V4 core, great value version, main frequency <= 48M                   |              |              |             |         |   |   |   |   |
| 1 = M3/ QingKe V3/V4 core, basic version, main frequency <= 72M                     |              |              |             |         |   |   |   |   |
| 2 = M3/ QingKe V4 non-floating-point core, enhanced version, main frequency <= 144M |              |              |             |         |   |   |   |   |
| 3 = QingKe V4F floating-point core, main frequency <= 144M                          |              |              |             |         |   |   |   |   |
| Device subfamily                                                                    |              |              |             |         |   |   |   |   |
| 03 = General-purpose                                                                |              |              |             |         |   |   |   |   |
| 05 = Connectivity (USB high-speed, SDIO, CAN)                                       |              |              |             |         |   |   |   |   |
| 07 = Interconnectivity (USB high-speed, CAN, Ethernet, DVP, SDIO, FSMC)             |              |              |             |         |   |   |   |   |
| 08 = Wireless (BLE5.X, CAN, USB, Ethernet)                                          |              |              |             |         |   |   |   |   |
| 35 = Connectivity (USB, USB PD/Type C)                                              |              |              |             |         |   |   |   |   |
| 33 = Connectivity (USB)                                                             |              |              |             |         |   |   |   |   |
| Pin count                                                                           |              |              |             |         |   |   |   |   |
| J = 8 pins                                                                          | A = 16 pins  | F = 20 pins  | G = 28 pins |         |   |   |   |   |
| K = 32 pins                                                                         | T = 36 pins  | C = 48 pins  | R = 64 pins |         |   |   |   |   |
| W = 68 pins                                                                         | V = 100 pins | Z = 144 pins |             |         |   |   |   |   |
| J = 8 pins                                                                          | A = 16 pins  | F = 20 pins  | G = 28 pins |         |   |   |   |   |
| Flash memory size                                                                   |              |              |             |         |   |   |   |   |
| 4 = 16 Kbytes of Flash memory                                                       |              |              |             |         |   |   |   |   |
| 6 = 32 Kbytes of Flash memory                                                       |              |              |             |         |   |   |   |   |
| 7 = 48 Kbytes of Flash memory                                                       |              |              |             |         |   |   |   |   |
| 8 = 64 Kbytes of Flash memory                                                       |              |              |             |         |   |   |   |   |
| B = 128 Kbytes of Flash memory                                                      |              |              |             |         |   |   |   |   |
| C = 256 Kbytes of Flash memory                                                      |              |              |             |         |   |   |   |   |
| Package                                                                             |              |              |             |         |   |   |   |   |
| T = LQFP                                                                            | U = QFN      | R = QSOP     | P = TSSOP   | M = SOP |   |   |   |   |
| Temperature range                                                                   |              |              |             |         |   |   |   |   |
| 6 = -40°C~85°C (industrial-grade)                                                   |              |              |             |         |   |   |   |   |

7 = -40°C~105°C (automotive-grade 2)

3 = -40°C~125°C (automotive-grade 1)

D = -40°C~150°C (automotive-grade 0)